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Supplementary Material for Recovery-Aware Physical Attribution of Load and Line Events From Sparse PMUs

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ABSTRACT This supplement documents the implemented recovery-aware history bank: its local filtering calculations, measurement map, publication rules and simulation contracts. It records the complete confirmatory outcomes, controlled severity-score/state-carry ablation and continuous single-state control. The operating-point signature and recovery-geometry studies keep their separate, known-onset contracts and denominators. All empirical results retain their original campaign and threshold contracts. Separate development diagnostics check covariance, physical construction and nonlinear prediction refinement. Earlier voltage-reconstruction, simultaneous-source and general diagnosability studies are preserved in the separately compiled Historical Technical Notes. An explicitly prospective section specifies an adaptive extension and its separate cyber interface; its small-model checks are not campaign results.

INDEX TERMS phasor measurement units, sequential diagnosis, physical state continuity, conditional filtering, reproducibility.

APPENDIX A IMPLEMENTED PREDICTOR AND CONDITIONAL UNCERTAINTY

This section specifies the local operators and conditional distributions used by the fixed continuous estimator. It distinguishes identities of that local model from approximations introduced by its finite history bank. The numerical checks below exercise these identities on controlled inputs; they do not recalibrate or rerun the reported evaluation.

A. AFFINE OPERATORS AT A CONSISTENT TOPOLOGICAL ANCHOR

For network hypothesis ℓ, let $(\bar{x}_\ell, \bar{u}_\ell)$ be the operating point determined by the prefix-fitted load and dispatch factors. Fix its differential coordinates while solving the algebraic constraints separately for each topology q:

$$g_{\ell,q}(\bar{x}_\ell, \bar{z}_{\ell,q}, \bar{u}_\ell) = 0. \tag{1}$$

The resulting point need not be a differential equilibrium after a line opening. Assume a nonsingular algebraic Jacobian and a fixed angle gauge. In the expressions below, f is the differential vector field after division by any nonunit differential mass entries. All derivatives are evaluated at the topology-specific point in (1); the network index is suppressed for readability. Define

$$\begin{aligned} c_q &= f_q(\bar{x}, \bar{z}_q, \bar{u}), \\ A_q^c &= f_x - f_z g_z^{-1} g_x, & B_q^c &= f_u - f_z g_z^{-1} g_u, \\ C_q^c &= H_x - H_z g_z^{-1} g_x, & D_q^c &= H_u - H_z g_z^{-1} g_u, \\ d_q^{el} &= H_q(\bar{x}, \bar{z}_q, \bar{u}) - H_0(\bar{x}, \bar{z}_0, \bar{u}). \end{aligned} \tag{2}$$

For the admitted scalar load direction $\delta u = La$, the continuous approximation is $\dot{\delta x} = A_q^c \delta x + c_q + B_q^c La$ and $\delta y = C_q^c \delta x + d_q^{el} + D_q^c La$. For a line, the topology selects the operator and the amplitude direction is zero. A nonzero pre-event state contributes through δx; d_q^{el} supplies the zero-deviation offset at the fitted anchor. This is a local expansion, so large excursions can still cause prediction error even when the algebraic solve and derivatives are accurate.

For a mode lasting h, compute the affine flow without

inverting A_q^c:

$$\mathcal{A}_q = \begin{pmatrix} A_q^c & c_q & B_q^c L \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$
$$\exp(h\mathcal{A}_q) = \begin{pmatrix} F_q(h) & \beta_q(h) & U_q(h) \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (3)$$

Thus $\beta_q(h) = \int_0^h \exp(A_q^c s) c_q \, ds$. If an admitted switch divides one report into an old-mode interval h_1 and a new-mode interval h_2, then

$$F = F_2 F_1, \qquad \beta = F_2\beta_1 + \beta_2, \qquad U = F_2 U_1 + U_2. \quad (4)$$

The physical state is continuous across this composition. The final-mode measurement map uses the consistent algebraic right limit. The predictor then appends phase history and the deterministic memory of the instrumental transformation. Scaling these coordinates and applying the same affine augmentation to (4) produces the stored $T_o, J_o, g_o, u_o, j_o, v_o$ operators. In their report ordering,

$$r_k = J_o\xi_k + j_o + v_o a + \epsilon_k,$$
$$\xi_{k+1} = T_o\xi_k + g_o + u_o a + \eta_k. \quad (5)$$

The measurement is scored before correcting the state for the next report. The constant startup discrepancy is applied to the observations; it does not replace the topology-dependent affine terms.

The construction has three steps. A static solve of (1) exports the network forcing, the PMU reference and their reduced Jacobians; the conditional builder forms the topology offsets, the mode flows and the one-report operators; and a final check verifies that the constructed reference matches the prefix-fitted operating point. The reproduction manifest preserves these files and their hashes. No event truth is used to select the startup operating point.

B. FOUR AMPLITUDE INTERVALS REPRESENT ONE PRIOR

Let $I = [-0.2, 0.2]$ be partitioned into four disjoint intervals I_j up to endpoints of zero probability. Write $Z_J(\mu, v) = \int_J \mathcal{N}(a; \mu, v)\, da$ for any interval J. The initial prior has the exact decomposition

$$p_0(a) = \sum_{j=1}^{4} \alpha_{j,0} \, \mathrm{TN}_{I_j}(a; \mu_0, v_0),$$
$$\alpha_{j,0} = Z_{I_j}(\mu_0, v_0)/Z_I(\mu_0, v_0). \quad (6)$$

The implementation assigns one history per source, proposed onset, restoration path, and interval. Each retains its own $(m_{h,j}, d_{h,j}, \mu_{h,j}, v_{h,j}, w_{h,j})$; there is no per-report collapse to a single amplitude distribution. The four interval histories use the same physical operators. They are neither quadrature nodes nor four separately linearized severity models.

For a fixed physical history with observations Y, let $Z_j(Y) = \int_{I_j} p(Y \mid a, h)\, \mathrm{TN}_{I_j}(a; \mu_0, v_0)\, da$. Then

$$p(Y \mid h) = \sum_j \alpha_{j,0} Z_j(Y), \qquad \alpha_j(Y) = \frac{\alpha_{j,0} Z_j(Y)}{p(Y \mid h)}. \quad (7)$$

Applying the local predictive integral sequentially gives the same identity. Identical linear operators and gains preserve the common untruncated Gaussian parameters for interval copies of the same path; their truncated moments and normalized weights differ. Subsequent beam pruning can remove interval histories, so equivalence with an unpartitioned bank is not guaranteed after pruning. The recorded discarded mass describes that approximation and is not a bound on all future decisions.

C. AMPLITUDE DEPENDENCE AND MIXTURE COMPRESSION

Condition on a retained history and the reports used so far. For $a \sim \mathrm{TN}_{[l,u]}(\mu, v)$, set $\sigma = \sqrt{v}$, $\alpha = (l-\mu)/\sigma$, $\beta = (u-\mu)/\sigma$, $Z = \Phi(\beta) - \Phi(\alpha)$, and $\lambda = [\phi(\alpha) - \phi(\beta)]/Z$. Its moments are

$$\bar{a} = \mu + \sigma\lambda,$$
$$V_a = v\left[1 + \frac{\alpha\phi(\alpha) - \beta\phi(\beta)}{Z} - \lambda^2\right]. \quad (8)$$

The truncated-moment routine evaluates the normalizer in the log domain and bounds small numerical excursions of the moments. The Gaussian parameters μ, v are therefore distinct from the conditional amplitude mean and variance.

For the affine conditional state model $\xi \mid a, h \sim \mathcal{N}(m_h + d_h a, P_h)$, total expectation and covariance give

$$\bar{\xi}_h = m_h + d_h\bar{a}_h,$$
$$\mathrm{Cov}(\xi, a \mid h) = d_h V_{a,h},$$
$$V_h = P_h + d_h V_{a,h} d_h^\top. \quad (9)$$

With normalized history weights ω_h, including the no-new-event history, the complete mixture moments would be

$$\bar{\xi} = \sum_h \omega_h \bar{\xi}_h,$$
$$P_{\mathrm{mix}} = \sum_h \omega_h \left[V_h + (\bar{\xi}_h - \bar{\xi})(\bar{\xi}_h - \bar{\xi})^\top\right]. \quad (10)$$

This identity applies within a network hypothesis and its common state coordinates. It does not justify averaging absolute rotor states between different network equilibria.

The continuous estimator uses the shared regime covariance P_ρ in place of P_h. It updates the affine means and amplitude distributions, and computes marginal diagonal state variances for diagnostics. At closure it carries $\bar{\xi}$, clears explicit histories, and records the trace of amplitude/mean spread omitted from the new single-history representation. It does not carry the complete covariance in (10), nor the uncertainty of the fitted startup context. Instrumental and discrepancy states remain in the carried augmented mean.

TABLE 1. Deterministic local-model checks. Maximum absolute errors over 72 scalar updates, five switch positions, and 12 joint Gaussian updates; each row uses its stated reference.

Check	Maximum error
Log evidence: partition	1.78×10^{-15}
Amplitude mean/variance	4.16×10^{-17}
Log evidence: quadrature	3.11×10^{-15}
Affine flow: ODE reference	1.11×10^{-16}
Joint mean/covariance	8.33×10^{-17}

For comparison, an exact conditional covariance recursion for the local linear model (5), with independent noises of covariances R, Q, is

$$\begin{aligned} S_h &= J_o P_h J_o^\top + R, \\ K_h &= T_o P_h J_o^\top S_h^{-1}, \\ P_h^+ &= T_o P_h T_o^\top + Q - K_h S_h K_h^\top . \end{aligned} \tag{11}$$

This is a diagnostic reference, not a description of a new executed continuous estimator. The fixed implementation solves the Riccati equation for the nominal operator in each discrepancy regime and uses that shared covariance for the admitted switched operators.

Sequential likelihood factorization conditions on past reports; it does not require marginal independence of the observations. The instrumental VAR/AR filter and the latent discrepancy AR state already model temporal dependence. Remaining covariance misspecification, linearization error, adaptive proposals, and compression can nevertheless affect the resulting weights. Their empirical calibration requires pre-update residual and coverage diagnostics, separately from the algebraic checks below.

D. DETERMINISTIC VERIFICATION AND ITS BOUNDARY

A deterministic check loads the implemented amplitude routine without running a campaign. It compares sequential one-interval and four-interval evidence and moments with direct scalar quadrature, including zero information and amplitudes on interval boundaries. An independent small switched linear system checks affine-flow composition against numerical integration. A linear Gaussian reference also compares the conditional affine-state recursion with a joint state–amplitude Kalman recursion for an untruncated Gaussian amplitude; the truncated case is checked separately by quadrature. All histories are retained in these checks. Source hashes, tolerances, and complete results accompany the generated table. These checks establish local-model identities on the stated inputs. They establish neither calibration of the pruned network bank nor robustness of its nonlinear physical approximation, and add no event outcomes to the evaluation cohorts.

APPENDIX B PREDICTIVE SUFFICIENCY AND DISTRIBUTIONAL FUSION

This note distinguishes a full posterior over an adequate augmented state from its mean. Let S_k include physical, instrument and discrepancy memory, the predictive mode, uncertain fixed parameters, dwell age, and any other summary needed to make the process Markov. Let H_k retain historical cause labels. Assume, in particular, that mode transitions and other predictive effects are represented in S_k, so the next-step kernel is conditionally independent of old cause labels given S_k:

$$p(S_{k+1}, Y_{k+1} \mid S_{0:k}, H_{0:k}, Y_{1:k}) = p(S_{k+1}, Y_{k+1} \mid S_k).$$

Then the marginal filtering distribution $\pi_k(ds) = p(ds \mid Y_{1:k})$ is predictively sufficient. For a measurable report set B, define $K_B(s) = p(Y_{k+1} \in B \mid S_k = s)$. Conditional independence and the tower rule give

$$\begin{aligned} p(Y_{k+1} \in B \mid Y_{1:k}) &= \mathbb{E}[K_B(S_k) \mid Y_{1:k}] \\ &= \int K_B(s)\, \pi_k(ds). \end{aligned}$$

This is a standard filtering identity, not a new result. Adequacy is substantive: omitted memory, mode, dwell age, uncertain parameters, or model mismatch invalidate the factorization unless represented in the state/kernel.

The posterior mean alone is generally insufficient. For example, with $Y_{k+1} = S_k + \epsilon$, $\epsilon \sim \mathcal{N}(0, R)$, the predictive variance is $\mathrm{Var}(S_k \mid y_{1:k}) + R$; a point estimate omits the first term. Under linear-Gaussian transitions and observations, a Gaussian filtering posterior is specified by its mean and covariance; both must be carried for exact prediction. A comparison of historical fusion with mean-only fusion therefore changes both distributional spread and, where applicable, mode or history uncertainty; it does not isolate lost historical labels.

For a mixture with components j, weights w_j, conditional means m_j and covariances P_j, the aggregate moments are

$$\begin{aligned} \bar{m} &= \sum_j w_j m_j, \\ P &= \sum_j w_j \{P_j + (m_j - \bar{m})(m_j - \bar{m})^\top\}. \end{aligned}$$

If a component also carries scalar severity A_j with mean μ_j, variance V_j, and conditional state representation $Z \mid A_j, j$ having mean $m_j + d_j(A_j - \mu_j)$ and covariance P_j, then its joint state/severity moments are

$$\mathrm{Cov}(Z, A) = \sum_j w_j \{d_j V_j + (m_j - \bar{m})(\mu_j - \bar{\mu})\},$$

$$\begin{aligned} \mathrm{Var}(Z) &= \sum_j w_j (P_j + d_j V_j d_j^\top) \\ &\quad + \sum_j w_j (m_j - \bar{m})(m_j - \bar{m})^\top, \\ \mathrm{Var}(A) &= \sum_j w_j \{V_j + (\mu_j - \bar{\mu})^2\}. \end{aligned}$$

Here $\bar{\mu} = \sum_j w_j \mu_j$. The joint moments require a common severity coordinate. Components with no severity, such as line hypotheses, use the state-only identity. Thus within-component severity correlation and between-history spread

both contribute. These identities describe moment-preserving compression only; the resulting Gaussian approximation need not preserve a multimodal predictive distribution.

Predictive filtering and retrospective attribution are distinct. A causal filter updates history weights using reports available at each time. A later proposal for an earlier onset is a smoothing/replay operation: it needs the stored pre-onset filtering distribution and must replace, rather than duplicate, the replayed suffix likelihood. The current filtering posterior alone does not in general recover that earlier distribution. A retrospective cause query requires the joint posterior $p(S_k, H_k \mid Y_{1:k})$; the marginal state posterior can suffice for prediction while being insufficient for historical attribution. This distinction matches the implementation's saved-state replay and its explicitly approximate outer model weights.

Finally, if a reconstructed hidden voltage $\widehat{V}$ is computed solely from already observed PMU data Y and fixed model/configuration θ, then it is a deterministic statistic $g(Y, \theta)$. Conditional on Y and θ, it supplies no additional independent evidence: appending it to the likelihood as an independent measurement would double-count data. It may still be a useful representation for a computation or diagnostic. If its construction uses additional measurements or uncertain parameters, those dependencies must instead be modeled jointly. This is the direct data-processing consequence of deterministic post-processing, not a claim that reconstruction has no computational value.

Finite-state enumerations check prediction before and after marginalizing history labels, joint cause queries, the mixture moments and deterministic post-processing. A Gaussian-mixture example also shows that moment matching can change predictive tails. These checks verify the stated identities and counterexamples; they do not validate the power-system estimator or quantify the accuracy of its compressed representations.

APPENDIX C HOW TO READ AND REPRODUCE THE CURRENT METHOD

This companion begins with the recovery-aware causal method used in the current article. Sections C–E give its execution order, local derivations, worked examples and simulation contracts. The separately compiled Historical Technical Notes preserve full diagnosability proofs, earlier state-reconstruction and batch-estimator studies, and secondary evidence. The earlier studies support particular components; they are not additional tests of the complete causal algorithm.

The article retains the physical model, the defining evidence integral, the recovery mechanism, the decision outputs and decisive results. This supplement supplies the intermediate calculations. The Historical Technical Notes develop functional observability, nuisance-profiled information and source ambiguity, and distinguish the implemented batch estimator $\mathcal{E}_{\mathrm{B}}$ from the proposed joint estimator $\mathcal{E}_{\mathrm{J}}$. Neither should be silently substituted for the causal physical history bank. The conditional Gaussian updates use standard filtering identities [1]; the derivation below specifies their role and the approximations made by the implemented bank.

A. INPUTS, OUTPUTS AND THE INFORMATION BOUNDARY

The inputs are successive reports from eight PMUs, the fixed physical and instrumental operators, the admitted load/line catalog and the fixed method configuration. The output is a timestamped sequence of detection, family/source and restoration messages, together with retained attribution masses. True ON/OFF times, true sources and generating severities belong to the evaluator. They are not inputs to the causal algorithm.

Three times must remain distinct: the latest observed report, the estimated physical onset, and the public emission time. Looking back to propose an onset is compatible with causality when the message is emitted only after the required reports arrive. It does not permit backdating the public alarm.

B. EXECUTION ORDER FOR ONE STREAM

1) **Load the fixed operators.** Retain voltage reference, oriented terminal currents, load input directions, switch modes and instrument memory. A line opening changes the network and observation operators. It is not represented by fitting a load amplitude.
2) **Fit the startup context.** Use the first 300 reports for operating factors, common reference and noise scale. For each network hypothesis, set the voltage/current discrepancy to the difference between the observed prefix mean and its reference prediction. Keep that offset fixed afterwards. Frequency and ROCOF use their empirical prefix centers. The prefix is not scored again as independent model-selection evidence.
3) **Predict the next report.** Propagate the retained histories and evaluate each network model's predictive density before correcting states or choosing new histories with the current report.
4) **Propose candidate onsets.** The fixed screen uses lags 0–18 reports within a ten-second stored state history. Each proposal starts from a stored state before its proposed onset, including any recovery already present. The bank does not enumerate every possible onset in the stored interval.
5) **Score histories.** For loads, integrate severity with the bounded prior and the local conditional Gaussian likelihood. For lines, score the admitted switched operator. Score a report before using that same report for its conditional state correction.
6) **Update and compress.** Update severity and conditional states, network weights, restoration children and the retained beam. Record discarded mass. At episode closure retain the augmented-state mixture mean; the omitted between-history spread is a documented approximation, not an exact posterior covariance.
7) **Publish and record.** Aggregate active mass M, cause mass Γ and ΔM. Apply the fixed detection, diagnosis

and restoration rules. Save emission time and the last report used, in addition to estimated onset and source. Continue propagation through OFF.

C. WHY REPLAYED REPORTS ARE NOT NEW EVIDENCE

Let a retained parent have log score $\ell_{0:k} = \ell_{0:\tau-1} + \ell^{old}_{\tau:k}$. Replacing its suffix by a newly proposed physical history gives

$$\ell^{new}_{0:k} = \ell_{0:k} - \ell^{old}_{\tau:k} + \ell^{new}_{\tau:k}. \tag{12}$$

Transition and prior terms must accompany their corresponding history. The rolling screen supplies a proposal, not an extra independent likelihood. Adding the new suffix without removing the old suffix would count the same reports twice. This bookkeeping statement does not make adaptive screening or beam pruning exact Bayesian inference.

D. FIXED NUMERICAL CHOICES

The input grid is 30 reports/s. The five network hypotheses are nominal, global $\pm 2\%$ and selected-branch $\pm 3\%$ paired R/X perturbations. Each hypothesis is fixed in time; the weights change. They do not identify a continuous impedance posterior.

The per-report birth and restoration hazards are $1 - \exp[-(1/60)/30]$ and $1-\exp[-(1/30)/30]$. Birth mass is divided equally between load and line families, then uniformly among the admitted sources within each family. Load severity has a zero-mean Gaussian prior with standard deviation 0.15, truncated to $[-0.2, 0.2]$ and integrated over intervals bounded by $-0.2, -0.1, 0, 0.1, 0.2$. After three reports, eligible active histories can create OFF children at either half-report position, inheriting state and severity. After likelihood normalization, the beam retains the union of the 96 highest-weight recent histories, 64 older histories and at least one per occupied source. The union is renormalized.

The internal observer confirms active-mass evidence at 0.99 for six reports and has a separate 15-report internal OFF clock. The public physical-bank publisher requires an internal pending episode and six reports of source-qualified active support above 0.99; its public OFF requires six reports of the diagnosed cell's joint cause-and-restoration mass above 0.99. The detailed source qualification, diagnosis and replacement rules are given below. These thresholds are implementation settings, not guarantees of probability calibration or false-alarm control. Nominal and mixture use their fixed publication paths, so their comparison does not isolate network averaging alone. The admission-rule ablation changes publication, not the recovery dynamics.

E. SOURCE QUALIFICATION AND PUBLIC EPISODE RULES

The source screen and the onset search are distinct. Source qualification uses a 30-report (one-second) conditional innovation vector v with 47 coordinates per report. Fixed normal-mode gains propagate the stored pre-window state and whiten the suffix. For each prepared response s_j and lag from 0 to 25 reports, let

$$\begin{aligned} g_j &= s_j^\top s_j, \qquad u_j = s_j^\top v, \qquad R = v^\top v, \\ \widehat{a}_j &= \mathrm{clip}(u_j/\max(g_j, 10^{-30}), -0.2, 0.2), \\ G_j &= 2\widehat{a}_j u_j - \widehat{a}_j^2 g_j, \\ F_j &= \frac{\max(G_j, 0)}{\max\{(R - G_j)/1409, 1\}}. \end{aligned} \tag{13}$$

Line columns use fixed amplitude one. The source score is the maximum over its columns; the threshold is the $1 - 0.001/936$ quantile of an $F_{1,1409}$ distribution, approximately 24.0115. This threshold and the variance floor are a fixed proposal/publication rule, not a calibrated global false-alarm probability. The statistic is never multiplied into the Bayesian history likelihood. This one-second qualification screen does not enlarge the separate birth search beyond its 0–18 report lags.

Define $b_c = 1$ for a line, and $b_c = \mathbf{1}\{F_c > 24.0115\}$ for a load. At each monitored report, the publisher updates consecutive-report counters for $M_c b_c \geq 0.99$, $\Gamma_c \geq 0.99$, and $\Gamma_c P(\mathrm{OFF} \mid c) \geq 0.99$. A false condition clears its counter. An internal alarm opens a pending episode. A public alarm then requires either six pending reports with $\sum_c M_c b_c \geq 0.99$ or an already accumulated six-report line-active counter. The latter permits line evidence collected before the internal pending message to count.

A public episode receives one diagnosis when its highest cause-mass cell has six consecutive reports above 0.99, or at the 15-report deadline after the public alarm. The output family maximizes summed cause mass; sources are ranked within that family. The diagnosed source becomes the episode owner. Six reports of that owner's joint cause-and-OFF mass above 0.99 emit a public restoration. These publication operations do not rescale M or reset the physical state.

The internal mixture lifecycle separately replaces an episode after six reports with mass at least 0.99 on active histories whose onset is later than its alarm (using the implementation's quarter-report timing guard). It withdraws an unconfirmed episode after the diagnosis delay when null mass has exceeded 0.99 for 15 reports. Its internal restoration requires cause mass at least 0.99 and conditional retired mass at least 0.99 for 15 reports. These terminal messages are propagated to the public episode if already emitted; an unconfirmed pending episode is recorded as rejected. Finalization marks any remaining public episode right-censored. The nominal comparator retains its own fixed internal lifecycle, so the nominal/mixture comparison is not an isolated model-averaging ablation.

APPENDIX D LOCAL CALCULATIONS AND WORKED EXAMPLES

A. COMPLETING THE BOUNDED-AMPLITUDE INTEGRAL

For one retained severity interval, write $e = sa + \epsilon$, with $\epsilon \sim \mathcal{N}(0, S)$, $S \succ 0$, and $a \sim \mathrm{TN}_{[l,u]}(\mu, v)$, $v > 0$. Here μ, v are parameters of the underlying untruncated Gaussian, not

the bounded distribution's moments. With $g = s^\top S^{-1} s$ and $q = s^\top S^{-1} e$, expansion gives

$$(e - sa)^\top S^{-1}(e - sa) + (a - \mu)^2/v = e^\top S^{-1} e + \mu^2/v - \mu'^2/v' + (a - \mu')^2/v', \quad (14)$$

$$v' = (v^{-1} + g)^{-1}, \qquad \mu' = v'(\mu/v + q). \quad (15)$$

Define $Z(\mu, v) = \Phi((u-\mu)/\sqrt{v}) - \Phi((l-\mu)/\sqrt{v})$. Integrating the last Gaussian square over $[l, u]$ yields

$$\begin{aligned}\log L = & -\tfrac{1}{2}[e^\top S^{-1} e + \log|S| + n_y \log(2\pi)] \\ & + \tfrac{1}{2}[\log(v'/v) + \mu'^2/v' - \mu^2/v] \\ & + \log Z(\mu', v') - \log Z(\mu, v). \end{aligned} \quad (16)$$

The posterior on that interval is $\mathrm{TN}_{[l,u]}(\mu', v')$. With $\alpha = (l - \mu')/\sqrt{v'}$, $\beta = (u - \mu')/\sqrt{v'}$, $Z' = \Phi(\beta) - \Phi(\alpha)$ and standard normal density ϕ, its mean is

$$\mathbb{E}[a \mid e] = \mu' + \sqrt{v'}\,\frac{\phi(\alpha) - \phi(\beta)}{Z'}. \quad (17)$$

Thus neither the untruncated mean μ' nor a best-fit severity can simply replace the integral. The implementation combines retained intervals using their weights. The identity is exact for this conditional Gaussian model; it does not remove linearization, shared-covariance or pruning error.

B. NUMERICAL EXAMPLE OF ONE INTERVAL

This dimensionless example illustrates the calculation; it is not an IEEE-39 simulation or a new performance result. Set $S = 1$, $s = 2$, $e = 1$, $\mu = 0$, $v = 1$ and $[l, u] = [0, 1]$. Then $g = 4$, $q = 2$, $v' = 0.2$ and $\mu' = 0.4$. Substitution in (16) gives $L \simeq 0.342688$ and $\mathbb{E}[a \mid e] \simeq 0.464941$. The different posterior mean follows from truncation. A separate numerical check compares the closed form against direct numerical integration, including the posterior mean. It checks algebra only and does not modify a fixed experiment.

C. CONDITIONAL STATE CORRECTION

Represent the conditional state mean as $m_h + d_h a$. Let $T_o, J_o, g_o, u_o, j_o, v_o$ denote the stored one-report transition and observation operators for switch o. The code forms $e = r_k - J_o m_h - j_o$ and $s = J_o d_h + v_o$. With shared covariance P_ρ in discrepancy regime ρ,

$$\begin{aligned} S_o &= R + J_o P_\rho J_o^\top, & K_o &= T_o P_\rho J_o^\top S_o^{-1}, \\ m_h^+ &= T_o m_h + g_o + K_o e, & d_h^+ &= T_o d_h + u_o - K_o s. \end{aligned} \quad (18)$$

To see the last line, apply the correction to a fixed amplitude:

$$T_o(m_h + d_h a) + g_o + u_o a + K_o(e - sa) = m_h^+ + d_h^+ a.$$

The severity distribution is updated by the same likelihood integral. Regime covariances come from nominal-operator Riccati equations. The physical history bank does not propagate an independent exact covariance for every switched history. At closure the mean is retained and the omitted mixture spread recorded.

D. RESTORATION IS NOT A STATE RESET

Consider the illustrative scalar system $\dot{x} = -x + u$, $y = x$, with $x(0) = 0$. Apply $u = a$ for $0 \le t < \kappa$ and restore $u = 0$ afterwards. Direct integration gives

$$x(t) = \begin{cases} a(1 - e^{-t}), & 0 \le t < \kappa, \\ a(1 - e^{-\kappa}) e^{-(t-\kappa)}, & t \ge \kappa. \end{cases} \quad (19)$$

Both expressions agree at κ. For $a = 1$ and $\kappa = 2$, $x(2) = 0.864665$ and $x(3) = 0.318092$ even though the intervention is OFF. Resetting the reference to equilibrium would discard that predictable recovery. This example explains state continuity; it does not claim that one exponential describes the multimode IEEE-39 response.

E. ACTIVE MASS, CAUSE MASS AND MODEL REWEIGHTING

For one model, suppose an illustrative cell has mass 0.60 in active histories and 0.30 in restored histories, with 0.10 elsewhere. Its active mass is $M_c = 0.60$, while its originating-cause mass is $\Gamma_c = 0.90$. Recovery can therefore remain attributable without declaring the input active. These assigned masses illustrate definitions, not calibrated probabilities from an experiment.

For two reports, adding and subtracting $\sum_\ell w_{\ell,k} M_{\ell,c,k-1}$ proves

$$\begin{aligned}\Delta M_c = & \sum_\ell w_{\ell,k}(M_{\ell,c,k} - M_{\ell,c,k-1}) \\ & + \sum_\ell (w_{\ell,k} - w_{\ell,k-1}) M_{\ell,c,k-1}. \end{aligned} \quad (20)$$

For example, unchanged conditional masses $(0.8, 0.2)$ and network weights changing from $(0.5, 0.5)$ to $(0.75, 0.25)$ move M_c from 0.50 to 0.65. The entire 0.15 change comes from model reweighting. A rising aggregate mass need not mean that each model found stronger within-model evidence.

APPENDIX E SIMULATION SCENARIOS AND REPRODUCTION ORDER

A. PLANT, MEASUREMENT AND PROVENANCE

The simulator is a Julia/PowerDynamics research model of the IEEE 39-bus system, not an officially validated equivalent of the competition reference record. Its contract fixes Julia 1.11.9, PowerDynamics 5.0.0, NetworkDynamics 1.3.0 and IEEE39 0.1.0. Initial states, effective parameter vectors, terminal map, reporting operator and stochastic model come from that snapshot. Nominal inventory files do not replace the effective operating point. The load law is the fixed impedance law.

The voltage buses are $\{2, 5, 6, 10, 19, 22, 29, 39\}$. Each current is an oriented branch-terminal measurement, not a nodal injection. BUS5 retains the provisional historical 5–6 branch, terminal 5, hypothesis; producer CT identity is unconfirmed and BUS39 remains partially validated. Execution uses full-precision configuration files rather than rounded tables. The competition reference record provided retrospective calibration context only.

Each physical parent is integrated continuously through its interventions. Differential states remain continuous at switching, while algebraic values are recomputed consistently. Reports use $t = k/30$ with endpoints included; the switch-aligned report uses the post-switch algebraic value. Solver steps are recorded separately and observations are not interpolated across a switch. The causal instrumental history is preserved through OFF.

B. CONTINUOUS DEVELOPMENT AND CONFIRMATION

The two development campaigns each contain three 80-s event parents with six interventions and one 80-s normal parent. Crossing the four parents with three R/X conditions gives 12 physical trajectories, 18 unique interventions and 54 event realizations per campaign. Both are development evidence in the current article. Repeated conditions are not independent event draws.

The confirmatory evaluation prespecifies 18 continuous parents, 27 persistent-line parents (nine lines, three repetitions), and 12 normal parents. The three R/X conditions produce $3(18+27+12) = 171$ intended physical trajectories; two instrumental-noise scales yield 342 observed records. The continuous parents contain $18 \times 6 = 108$ unique interventions. Their six condition/noise realizations are repeated observations of those parents. The 120-s and 600-s grids contain 3601 and 18001 reports, respectively. Completed results appear in the dedicated confirmation tables.

Continuous evaluation pairs an alarm temporally before inspecting its predicted label. The first unmatched post-ON alarm is eligible up to the next ON, with the inherited 30-s cap. Timely attribution requires correct family and asset emitted before physical OFF. Unmatched alarms, false restorations and open episodes are retained. Persistent-line evaluation instead scores the first stopping decision, including early stops, wrong sources, omissions and censoring. The detector of Chen *et al.* is evaluated only there; it has no load or restoration output. Confidence intervals use 10,000 paired parent-cluster resamples, not independent PMU samples.

C. SEPARATE SIGNATURE AND RECOVERY CONTROLS

The context-signature study uses a single load at buses $\{3, 7, 16, 24\}$, a 10-s record, known ON at 3 s and OFF at 6 s, and one scalar operating-context direction. It contains 57 dictionary/check trajectories and 85 test trajectories (80 interventions and five controls), with three common noise seeds producing 255 observed test records. The context and severity are estimated from measurements, while onset and family are supplied. The generating source is used to assess the response approximation and severity error. Those conditional metrics do not measure unknown-onset causal attribution.

The recovery-geometry control uses 38 trajectories. Prior load histories at buses 7 and 24, followed by delays of 0.5 and 2 s, give four recovery contexts. Three source-pair comparisons within each context give 12 correlated comparisons. The prior input is active from 2 to 3 s; the subsequent response is examined over 1.5 s. The supplied history, onset, family and two-candidate universe define an oracle control, with one uncertain prior-severity direction. It does not evaluate the causal physical method, unknown onset, event cardinality or operational false alarms.

D. REPRODUCTION WITHOUT CHANGING THE EXPERIMENT

1) Read each campaign's fixed plan and manifest. Select the intended population before interpreting any denominator. Keep current causal, signature, recovery-control and historical batch studies separate.
2) Resolve the plant state, parameter and measurement files from the versioned snapshot. Every new simulation is a new run, and a new hypothesis a new candidate; earlier preparation steps are never rerun inside a fixed baseline.
3) Preserve ON/OFF continuity, report endpoints, noise seeds and the boundary between observations and reserved event truth. Run inference before releasing truth to the evaluator.
4) Retain solver/inference failures in intended denominators. Aggregate at the declared parent level and preserve the original comparators and publication rules. Do not retune from confirmation results.
5) Rebuild the two documents from the versioned tables and figures. The default build does not regenerate scientific artifacts or import new campaign results; refreshing artifacts is a separate operation.

The reproducibility package keeps this walkthrough, the campaign evidence and the earlier mathematical development, compiled separately as Historical Technical Notes. A successful PDF build verifies document consistency; it is not a new scientific validation.

APPENDIX F EXECUTED MEASUREMENT MAP

A. MEASUREMENT CONTRACT

The executed channel map is taken from the versioned research-plant snapshot. The eight voltage buses are $\{2, 5, 6, 10, 19, 22, 29, 39\}$. The internal voltage phasor is in per unit with unit gain and is exported as phase-to-neutral RMS volts using 345 kV/$\sqrt{3}$; voltage angle uses one common affine phase reference plus the stored relative offsets. The reference interpretation is observational: physical common drift and instrument clock drift are not separately identified. Terminal currents are converted to outward branch convention from the simulator's into-terminal observable, then mapped with the row-specific polarity, gain and phase bias. Current bases use $100\,\mathrm{MVA}/(\sqrt{3}\,V_{\mathrm{terminal}})$; the common voltage base is 199.186 kV.

Frequency is computed as $60 + \dot{\theta}/(360)$ from the reported voltage angle, and ROCOF as $\ddot{\theta}/360$ using a causal 21-report quadratic estimate. The generator fit is robust and magnitude weighted; the predictor uses a fixed unweighted quadratic fit over the same history. Four warmup outputs are unavailable; the confirmatory evaluation uses a 10-s startup context and

TABLE 2. Eight PMU rows of the executed measurement map. V is a bus voltage in p.u.; I is an oriented branch-terminal phasor. Gains and phase biases are applied to current channels. BUS5 retains the historical 5–6, terminal-5 hypothesis and its producer CT identity is unverified. BUS39 current phase is partially validated.

PMU	V bus	Edge	Branch	Terminal	Side	Polarity	Gain	Bias (°)	I unit	Provenance
2	2	2	2–3	3	dst	1	0.9581	-1.173	A	Terminal hypothesis; CT unverified
5	5	9	5–6	5	src	1	1.0361	0.312	A	BUS5 5–6 historical hypothesis; CT unverified
6	6	11	6–7	6	src	1	1.0077	1.495	A	Terminal hypothesis; CT unverified
10	10	17	10–11	11	dst	-1	1.0041	2.224	A	Terminal hypothesis; CT unverified
19	19	26	16–19	19	dst	1	1.0384	0.481	A	Terminal hypothesis; CT unverified
22	22	34	21–22	22	dst	1	1.0189	0.094	A	Terminal hypothesis; CT unverified
29	29	44	28–29	29	dst	-1	1.0316	-6.170	A	Terminal hypothesis; CT unverified
39	39	1	1–39	39	dst	-1	0.9298	30.053	A	BUS39 partial phase validation

TABLE 3. Fitted ExtraTrees comparators used in the held-out streams. Both were trained for this study and are distinct from the original competition models.

Comparator	Features	Windows	Trees
ExtraTrees	846	1, 5, 10 s	400
ExtraTrees short	1457	1, 5, 10 s; 3, 8, 15 reports	400

begins monitoring after 12 s. The 47 retained coordinates exclude the BUS39 current-angle coordinate. Noise multipliers 1 and 2 scale the standard deviation under the fixed stream. The stochastic model stores a cross-signal covariance field, but the executed report operator applies noise by signal family without that cross-family factor. These choices describe a simulated sensor configuration, not a verified eight-PMU CT/PT installation.

APPENDIX G LEARNED COMPARATOR LINEAGE AND DECISION CONTRACT

Both models were fitted on 32 physical parents from four earlier development campaigns; validation used 8 parents and 16 observed records from a fifth campaign. All of these were development data for the comparator construction. The fitted label set contains normal operation and all 18 catalog sources. The training target is the active source; after restoration, including physical recovery, its label is normal. This is not a source-excluded transfer test.

Each report selects the largest tree vote, and six consecutive reports of the same label drive the episode interface; there is no separate probability threshold. Tree votes are uncalibrated; the 300-report prefix and the start of monitoring at report 360 are fixed. Model hashes, training metadata and the observed class set are kept in the reproducibility package.

APPENDIX H COMPLETE CONFIRMATORY OUTCOMES AND DIAGNOSTIC AUDIT

The fixed test contains 171 physical trajectories and 342 observed records. Every method has an accounted output for every intended record; the solver-rejection and inference-failure ledgers are empty. The continuous, persistent-line and normal physical-parent counts are 18, 27 and 12. Three network conditions and two noise levels repeat these parents. Normal controls are shared between evaluation lanes, not doubled independent exposure. Tables below are generated directly from the stored summary, records, event matches and paired intervals.

A. TIMELY COVERAGE VERSUS CLASSIFICATION

The additional confusion view in Fig. 1 is a posthoc descriptive audit. It uses the original evaluator's temporal episode association and the first public diagnosis; it does not rematch events using source labels. A diagnosis emitted after physical OFF is counted as unavailable for this view. All diagnoses that are timely and matched in these four procedures have the correct family and exact source. The large load gap therefore reflects whether a diagnosis is available before OFF, not a measured advantage in conditional label accuracy among timely outputs. Unmatched alarms remain in the separate alarm tables. Normal records are not mixed into this event-level matrix.

B. MAINTAINING A SUSTAINED EVENT AFTER THE FIRST STOP

The first-stop score alone hides subsequent fragmentation. Although both physical banks and both learned comparators correctly identify all 162 persistent openings first, the learned comparators later return to an inactive label and issue additional alarms. Table 11 describes those already recorded outputs. It does not change the fixed first-stop outcome or extend Chen's one-stop contract into a recurrent monitor. Retained open episodes are appropriate when an intervention continues beyond the record.

C. MEASUREMENT INTEGRITY IS A SEPARATE EXECUTED COMPONENT

The availability interface correctly identifies all four missing-data intervals at the first received report that reveals the loss. This is a received-report clock, not zero transport or computation latency. Bad-data labels come from the inherited window classifier. Neither missingness nor corruption proves malicious intent. The composition preserves physical-source and affected-PMU identities but does not run a joint physical/cyber Bayesian filter; the physical bank is unavailable on the retrospective replay of the competition reference record under its startup noise-scale contract.

TABLE 4. Continuous outcomes by network and noise stratum. Each cell is timely family/source decisions / intended events; unmatched event-stream alarms. Replays share physical parents.

Network	Noise	Physical bank	Nominal bank	ExtraTrees	ET short
Matched	1	108/108; 1	108/108; 1	56/108; 6	60/108; 5
Matched	2	108/108; 0	108/108; 0	54/108; 4	58/108; 0
In-bank	1	108/108; 2	108/108; 3	56/108; 5	60/108; 5
In-bank	2	108/108; 1	108/108; 2	54/108; 4	57/108; 0
Out-of-bank	1	106/108; 38	105/108; 47	56/108; 5	60/108; 6
Out-of-bank	2	106/108; 30	106/108; 39	54/108; 5	57/108; 0

TABLE 5. Continuous monitoring: unavailable intended records, normal-record alarm incidence, normal alarm counts and available normal exposure, false restorations and open episodes. Counts combine six repeated strata descriptively; normal parents are not independent across strata. Missing outputs remain intended failures.

Method	Unavailable	Normal incidence	Normal alarms	Exposure (h)	False OFF	Open at end
Physical 5-model	0	0/72	0	11.760	58	0
Physical nominal	0	0/72	0	11.760	68	0
ExtraTrees	0	0/72	0	11.760	29	0
ET short	0	0/72	0	11.760	16	0

TABLE 6. Conditional median diagnosis/restoration delays (s), kept within each paired stratum. These medians summarize matched emitted diagnoses/restoration messages with finite delay and must be read alongside omissions, unavailable records and censoring; – denotes no finite estimate. Logical delays do not measure computational throughput.

R/X condition	Noise	Physical 5-model	Physical nominal	ExtraTrees	ET short
Matched	1	0.27/0.18	0.27/0.18	0.23/0.18	0.19/0.18
Matched	2	0.27/0.18	0.27/0.18	0.30/0.18	0.20/0.18
In-bank	1	0.27/0.18	0.27/0.18	0.23/0.18	0.19/0.18
In-bank	2	0.27/0.18	0.27/0.18	0.30/0.18	0.20/0.18
Out-of-bank	1	0.33/0.18	0.33/0.18	0.23/0.18	0.19/0.18
Out-of-bank	2	0.26/0.18	0.33/0.18	0.30/0.18	0.20/0.18

TABLE 7. Persistent-line comparison on the common observations. Each cell gives correct first source / intended openings; pre-change stops / right-censored openings; mean restricted correct delay (s). Wrong first sources remain in the denominator. Restricted delay retains the horizon penalty for failures and censoring defined by the fixed evaluator.

R/X condition	Noise	Physical 5-model	Physical nominal	Chen *et al.*
Matched	1	27/27; 0/0; 0.18	27/27; 0/0; 0.18	12/27; 11/0; 54.58
Matched	2	27/27; 0/0; 0.18	27/27; 0/0; 0.18	12/27; 11/0; 54.58
In-bank	1	27/27; 0/0; 0.18	27/27; 0/0; 0.18	13/27; 11/0; 51.04
In-bank	2	27/27; 0/0; 0.18	27/27; 0/0; 0.18	13/27; 11/0; 51.05
Out-of-bank	1	27/27; 0/0; 0.18	27/27; 0/0; 0.18	12/27; 11/0; 54.58
Out-of-bank	2	27/27; 0/0; 0.18	27/27; 0/0; 0.18	12/27; 11/0; 54.59

TABLE 8. Persistent-line coverage and stopping outcomes across six repeated strata, including both fixed ExtraTrees comparators. Normal controls are the same controls used in the continuous table, not additional exposure. Chen *et al.* normal exposure ends at its first stop; the other methods monitor the full available record.

Method	Unavailable	Correct first	Pre-change	Censored	Normal incidence	Exposure (h)
Physical 5-model	0	162/162	0	0	0/72	11.760
Physical nominal	0	162/162	0	0	0/72	11.760
ExtraTrees	0	162/162	0	0	0/72	11.760
ET short	0	162/162	0	0	0/72	11.760
Chen *et al.*	0	74/162	66	0	72/72	0.701

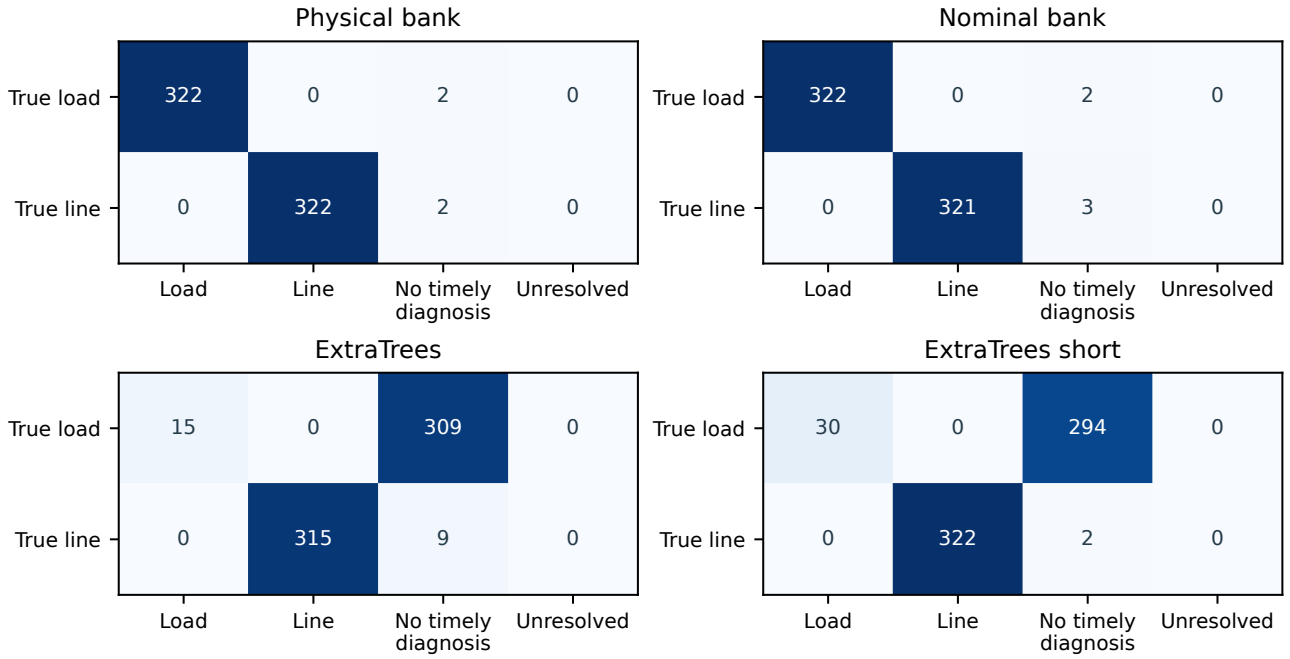


FIGURE 1. **Posthoc confusion audit of the fixed continuous test, with 324 realizations per true family. No timely diagnosis includes omitted and late diagnoses. Counts reuse the registered episode matches; all 432 prediction JSON files used were checked against the stored prediction hashes. These panels do not include unmatched alarms or constitute a new predeclared endpoint.**

TABLE 9. **Timely decisions by source in the continuous test. These are correlated realizations of the registered physical schedules, not source-excluded generalization.**

Family / asset	Physical bank	Nominal bank	ExtraTrees	ET short
Line / LINE14-15	36/36	36/36	27/36	34/36
Line / LINE16-17	36/36	36/36	36/36	36/36
Line / LINE17-18	48/48	48/48	48/48	48/48
Line / LINE23-24	30/30	30/30	30/30	30/30
Line / LINE26-27	24/24	24/24	24/24	24/24
Line / LINE26-29	24/24	23/24	24/24	24/24
Line / LINE3-4	42/42	40/42	42/42	42/42
Line / LINE4-14	46/48	48/48	48/48	48/48
Line / LINE9-39	36/36	36/36	36/36	36/36
Load / BUS15	42/42	42/42	6/42	6/42
Load / BUS16	42/42	42/42	0/42	3/42
Load / BUS20	12/12	12/12	6/12	12/12
Load / BUS24	42/42	42/42	0/42	0/42
Load / BUS26	18/18	18/18	0/18	0/18
Load / BUS3	42/42	42/42	3/42	3/42
Load / BUS31	40/42	40/42	0/42	0/42
Load / BUS4	36/36	36/36	0/36	6/36
Load / BUS7	48/48	48/48	0/48	0/48

TABLE 10. Paired physical-bank minus learned-comparator differences. Timely attribution is in percentage points; extras are per monitored event-record hour, excluding 12 s of startup from each 120-s record. The registered rate differences and 95-percent parent-bootstrap bounds are exactly rescaled by 120/108; original counts, matches and intervals remain stored.

Network	Noise	Method	Reference	Timely difference	Extra-rate difference
Matched	1	Physical bank	ExtraTrees short	44.44 [40.48, 47.92]	-7.41 [-16.67, 0.00]
Matched	1	Physical bank	ExtraTrees	48.15 [44.44, 51.67]	-9.26 [-20.37, 0.00]
Matched	1	Nominal bank	ExtraTrees short	44.44 [40.74, 48.04]	-7.41 [-14.81, 0.00]
Matched	1	Nominal bank	ExtraTrees	48.15 [44.44, 51.67]	-9.26 [-20.37, 0.00]
Matched	2	Physical bank	ExtraTrees short	46.30 [42.75, 49.12]	0.00 [0.00, 0.00]
Matched	2	Physical bank	ExtraTrees	50.00 [46.30, 53.70]	-7.41 [-14.81, -1.85]
Matched	2	Nominal bank	ExtraTrees short	46.30 [42.86, 49.12]	0.00 [0.00, 0.00]
Matched	2	Nominal bank	ExtraTrees	50.00 [46.38, 53.70]	-7.41 [-14.81, -1.85]
In-bank	1	Physical bank	ExtraTrees short	44.44 [40.62, 47.92]	-5.56 [-14.81, 5.56]
In-bank	1	Physical bank	ExtraTrees	48.15 [44.44, 51.67]	-5.56 [-14.81, 5.56]
In-bank	1	Nominal bank	ExtraTrees short	44.44 [40.62, 47.92]	-3.70 [-12.96, 7.41]
In-bank	1	Nominal bank	ExtraTrees	48.15 [44.44, 51.67]	-3.70 [-12.96, 5.56]
In-bank	2	Physical bank	ExtraTrees short	47.22 [43.18, 50.98]	1.85 [0.00, 5.56]
In-bank	2	Physical bank	ExtraTrees	50.00 [46.49, 53.70]	-5.56 [-11.11, 0.00]
In-bank	2	Nominal bank	ExtraTrees short	47.22 [43.33, 51.04]	3.70 [0.00, 9.26]
In-bank	2	Nominal bank	ExtraTrees	50.00 [46.30, 53.70]	-3.70 [-11.11, 3.70]
Out-of-bank	1	Physical bank	ExtraTrees short	42.59 [38.60, 46.43]	59.26 [40.74, 79.63]
Out-of-bank	1	Physical bank	ExtraTrees	46.30 [42.11, 50.00]	61.11 [40.74, 81.48]
Out-of-bank	1	Nominal bank	ExtraTrees short	41.67 [36.11, 46.67]	75.93 [51.85, 100.00]
Out-of-bank	1	Nominal bank	ExtraTrees	45.37 [41.11, 50.00]	77.78 [50.00, 105.56]
Out-of-bank	2	Physical bank	ExtraTrees short	45.37 [41.11, 50.00]	55.56 [38.89, 72.22]
Out-of-bank	2	Physical bank	ExtraTrees	48.15 [43.59, 52.50]	46.30 [27.78, 66.67]
Out-of-bank	2	Nominal bank	ExtraTrees short	45.37 [41.11, 50.00]	72.22 [46.30, 98.15]
Out-of-bank	2	Nominal bank	ExtraTrees	48.15 [43.75, 52.56]	62.96 [35.19, 88.89]

TABLE 11. Descriptive persistence audit after first detection on sustained line openings. All four procedures first identify 162/162 lines. Extras are later unmatched alarms; unmatched OFF counts only restoration messages on unmatched episodes. Open-at-end messages are expected for sustained events and are not automatically errors. False-negative reports include detection delay and later inactive labels. Chen is a first-stop method and is excluded from this recurrent-monitoring diagnostic.

Method	Openings	Extras	Unmatched OFF	Open at end	FN reports
Physical bank	162	0	0	162	810
Nominal bank	162	0	0	162	810
ExtraTrees	162	164	164	16	335374
ExtraTrees short	162	179	178	61	182600

TABLE 12. Retrospective measurement-integrity components over 180 windows. Hits require family and PMU identity. Missingness is observed availability; corrupted-data identification is inherited from the window classifier. These are not counts of attributed cyberattacks.

Component	Positive-window hits	FP	FN
Missing	14/14	0	0
Corrupt	1/2	2	1

APPENDIX I HELD-OUT RECOVERY COMPARISON PROTOCOL

The three arms use one causal reporting clock. The two physical arms share the nominal operators and permitted startup prefix. The continuous GLR retains one committed state and considers transition alternatives over a causal 19-report search; the bank retains conditional histories. ExtraTrees uses causal PMU features and was trained on a disjoint split with all catalog sources represented.

Selection used CAL only: among thresholds satisfying the normal alarm budget, maximize timely joint attribution, then minimize extra alarms and delay. The continuous GLR was rerun for each candidate threshold because commitment changes its state. TEST was evaluated once, after all predictions had been fixed. Timing associations use alarm time, not the true class. Timely attribution requires the right family and source before physical restoration; unavailable cases count in the intended denominator. Event-record extras and normal-record alarms have separate monitored exposures.

A. COMPARISON AT A COMMON CALIBRATION BUDGET

On a separate cohort, the nominal bank and the continuous generalized likelihood ratio (GLR) observer used the same nominal operators, eight PMUs, source catalog and permitted startup prefix. The GLR carries one committed state through restoration. ExtraTrees was refitted on disjoint simulated parents covering the same catalog. A common publication clock and calibration procedure selected a threshold for each arm under a normal-alarm budget of one per hour. The fixed methods were then scored once on 18 event parents and 24 normal parents. Interventions did not overlap, and the next onset was 3–8 s after physical restoration.

Figure 2 shows a 25.0 percentage point gain over GLR (paired 95% CI [13.89, 37.04]), with fewer extra alarms. The gain over ExtraTrees is 71.3 points (95% CI [63.89, 78.70]), with more extra alarms. The intervals resample entire physical parents within the network/noise strata; events within a parent do not count as independent samples. All three methods had zero normal alarms in 3.92 monitored hours per method (one-sided 95% Poisson upper bound: 0.76/h). The result is conditional on the count model and the aggregate exposure. The two physical methods differ in candidate search, score semantics and commitment, so the comparison does not isolate the effect of explicit recovery histories.

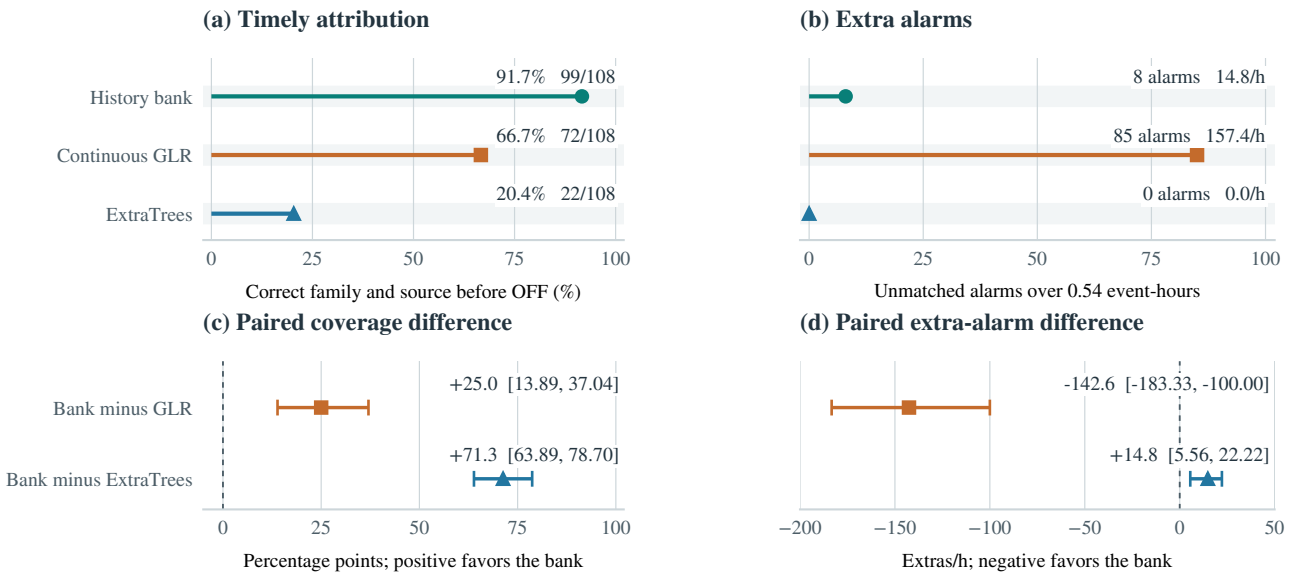
A separate calibration study found no gain from its chosen covariance correction: its selected correction multiplier was zero, and all three calibration arms tied on the evaluated endpoints. The profile and marginal scores likewise tied in the fixed-threshold carry/reset study. Neither finding establishes equivalence beyond its own comparison.

APPENDIX J CONTROLLED COMPONENT ABLATION

The fixed protocol and all output tables are part of the reproducibility package. Each of four arms accounts for all 24 intended records (18 continuous, three normal and three persistent): the evaluation audit records 96/96 available outputs and zero failures. The trial is posthoc and no score threshold is calibrated to these outcomes. Each normal control is a 600-s record with 588 s of evaluated exposure after startup, for 1,764 s total.

a: Exposure convention for the ablation alarm rate.

The ablation table reports extras per event-hour with the complete 120-s event record, including startup, as denominator. Its continuous exposure is $18 \times 120 = 2160$ s. The main article's confirmatory alarm-rate figure instead uses monitored exposure after the 12-s startup. Under that convention the ablation exposure would be $18 \times 108 = 1944$ s and each reported rate difference and interval bound would be



Normal records: 0 alarms in 3.92 h per method; one-sided 95% Poisson upper bound 0.76/h.
Paired 95% intervals: 10,000 resamples of 18 event parents. Each method meets the calibration alarm budget.

FIGURE 2. **Coverage and alarm cost on held-out sequences. (a) Correct family and source before OFF, out of 108 interventions. (b) Unmatched alarms over 0.54 monitored event-hours. (c,d) History bank minus each comparator, with 95% intervals from 10,000 paired resamples of 18 event parents. Positive coverage differences and negative alarm-rate differences favor the bank. The bank improves both outcomes over GLR but trades more extras for higher coverage than ExtraTrees. Thresholds were selected on separate calibration data under a target of one normal alarm/h; normal exposure and its Poisson upper bound are shown separately.**

TABLE 13. **All outcomes, including restoration messages on unmatched episodes and episodes open at the record end.**

method	lane	available	events	timely	extras	false restorations	open eof
Marginal + carry	continuous	18	108	108	1	1	0
Marginal + carry	normal	3	0	0	0	0	0
Marginal + carry	persistent line	3	3	3	0	0	3
Marginal + reset	continuous	18	108	75	166	130	11
Marginal + reset	normal	3	0	0	0	0	0
Marginal + reset	persistent line	3	3	3	0	0	3
Profile + carry	continuous	18	108	108	1	1	0
Profile + carry	normal	3	0	0	0	0	0
Profile + carry	persistent line	3	3	3	0	0	3
Profile + reset	continuous	18	108	75	166	130	11
Profile + reset	normal	3	0	0	0	0	0
Profile + reset	persistent line	3	3	3	0	0	3

multiplied by $120/108$. Counts, event matches and the stored table are unchanged; these two rate conventions must not be compared without this conversion.

APPENDIX K EXECUTABLE SINGLE-STATE COMPARISON

This control uses the same conditional nominal plant and measurement operators as the nominal history bank, but retains only one committed mode and its augmented conditional state. Temporary alternatives live for at most 19 reports. The five-model history bank additionally mixes resistance and reactance alternatives. Proposals begin at report index 300 and publication at index 360, with zero-based indices. Each available report after proposal startup starts two subframe transition candidates for every admitted source when inactive, or two OFF candidates for the committed source when active. Every candidate uses the covariance regime chosen from preceding reports; the current sample cannot select its own predictive covariance.

For a load ON alternative, write the innovation as $e_k - as_k$. The affine conditional state is $m_k + ad_k$. Accumulate over its contiguous suffix

$$G = \sum_k s_k^\top S_k^{-1} s_k, \tag{21}$$

$$U = \sum_k s_k^\top S_k^{-1} e_k, \tag{22}$$

$$C = -\frac{1}{2} \sum_k \left(e_k^\top S_k^{-1} e_k + \log |S_k| + p \log(2\pi) \right). \tag{23}$$

Then $\widehat{a} = \mathrm{clip}(U/G, -0.2, 0.2)$ when $G > 0$, with $\widehat{a} = 0$ for a flat objective. The profile contrast is

$$\Lambda = C + \widehat{a}U - \tfrac{1}{2}\widehat{a}^2 G - C_{ref}. \tag{24}$$

TABLE 14. Paired descriptive differences with 95-percent physical-parent bootstrap intervals; no multiplicity-adjusted superiority claim.

method	reference	metric	difference	lower95	upper95
Marginal + carry	Profile + carry	timely fraction	0.000	0.000	0.000
Marginal + carry	Profile + carry	extras per event-hour	0.000	0.000	0.000
Marginal + carry	Marginal + reset	timely fraction	0.306	0.241	0.380
Marginal + carry	Marginal + reset	extras per event-hour	-275.000	-325.000	-223.333
Marginal + reset	Profile + reset	timely fraction	0.000	0.000	0.000
Marginal + reset	Profile + reset	extras per event-hour	0.000	0.000	0.000
Profile + carry	Profile + reset	timely fraction	0.306	0.241	0.380
Profile + carry	Profile + reset	extras per event-hour	-275.000	-325.000	-223.333

Here C_{ref} is the sum of the committed predictor's pre-update conditional log densities on exactly the same suffix. A line uses its discrete switched operator, without fitting a load amplitude. The reference uses its preceding severity estimate for prediction and can refine that estimate after receiving the new report. A candidate never changes the reference state before commitment. A load OFF alternative holds its birth amplitude fixed rather than re-profiling it.

ON scores subtract $\log(18 \times 2 \times 19)$; OFF scores subtract $\log(2 \times 19)$. These fixed search penalties are heuristics. Maximizing over overlapping suffixes and bounded amplitudes does not produce a calibrated posterior or a sequential false-alarm guarantee. For the history banks, ON uses the log odds of $M(c)$ and OFF the log odds of $M_{cause}(c)P(OFF \mid c)$.

The common clock requires the same winning ON source to exceed $\log 99$ for six consecutive monitored reports. It emits alarm and source diagnosis together. Six qualifying OFF reports for that source close the episode. A clock-induced commitment selects the candidate's current conditional state. Its estimate may change through hypothesis selection; it is never replaced with equilibrium or a zero state. A missing report propagates the retained mode without measurement correction, clears the short alternatives and contributes no confirmation evidence. The development comparison contains no such post-startup missing observations.

After OFF, the preceding discrepancy regime persists until 60 consecutive available, inferred-inactive reports meet the inherited quiet criterion. Its innovation is computed with the normal operator from the preceding committed state. A new active episode or missing report clears that quiet run. Retrospective source labels and event times never enter these steps.

The comparison has one fixed publication clock but different evidence semantics, search and severity treatment. It therefore compares complete procedures. It does not establish a matched false-alarm operating point, nor isolate a causal effect of long-history averaging. The source tree, execution hashes, every prediction and the successor event metrics are preserved in the experiment manifest.

APPENDIX L DEVELOPMENT STREAMS AND ADMISSION CONTROL

A. TIMELY ATTRIBUTION AND EXTRA ALARMS MEASURE DIFFERENT OUTCOMES

In the second development campaign, both ExtraTrees adapters and both physical references identify all tested lines in time (Table 16). The difference is in loads: the five-model bank obtains 26/27 timely decisions, nominal 27/27, and these ExtraTrees adapters 0/27. This tests transfer of the fixed adapters; it neither rules out supervised load detection nor reproduces the historical multi-class hackathon score.

TABLE 15. Open-development comparison with the same six-report publication clock. All 54 event replays are retained. Threshold score semantics differ; this does not establish equal false-alarm operating points. Missing closure includes missed events.

Procedure	Timely	Extras	Early OFF	No OFF
Single state, short GLR	37/54	37	0	4
Nominal history bank	44/54	6	0	10
Five-model history bank	51/54	3	0	3

TABLE 16. What the causal component resolves in the second development campaign. Load and line columns require the correct family and source emitted before OFF. The 54 realizations repeat 18 interventions across three R/X conditions; they are not 54 independent trials. Normal exposure is only 204 s per method.

Method	Load timely	Line timely	Extras	Normal alarms
Physical (5 models)	26/27	27/27	4	0
Physical nominal	27/27	27/27	7	0
ExtraTrees ref.	0/27	27/27	3	0
ET short	0/27	27/27	3	0
Onset witness	25/27	27/27	0	0

The five-model bank detects and eventually attributes 54/54 realizations, but only 53 diagnoses precede physical OFF. Nominal has 54 timely diagnoses, with 7 extras versus 4. Detection, eventual attribution and timely attribution therefore give different answers. The configured five-model procedure yields one fewer timely diagnosis and three fewer extra alarms than nominal; this comparison does not isolate model uncertainty.

B. RECOVERY AND A NEW EVENT MUST REMAIN COMPETING EXPLANATIONS

Figure 4 shows what the counts omit. The physical state continues evolving after OFF. Retained cause/recovery histories explain that trajectory without resetting the network; a later source must compete against both no-new-event recovery and other interventions.

The out-of-bank BUS31 case shows a remaining timing failure: the source is recovered eventually but too late. The

network mixture changes how much mass is assigned to that source, and the mass-increment identity in the main article separates that revision from within-model evidence. This decomposition distinguishes changing within-model evidence from changing network weights; it is an interpretation of the delayed decision.

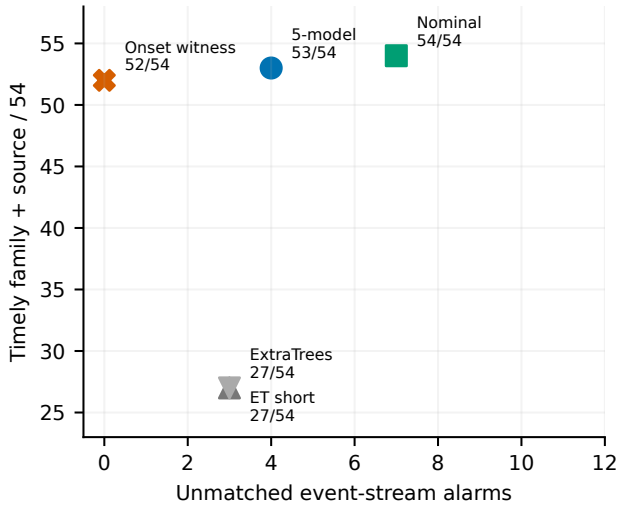


FIGURE 3. **The decision that an accuracy-only table hides. Development outcomes trade timely attribution against unmatched event-stream alarms. The onset-witness ablation removes all extras but loses a real BUS16 intervention; it is retained as negative evidence. Labels use 54 paired realizations of 18 interventions and do not imply 54 independent trials.**

The onset-witness ablation obtains zero extras but only 52/54 timely attributions: weak early evidence rejects a real BUS16 load. It inherits the physical states and masses, so suppressing the alarm does not repair the underlying explanation. This negative result left the five-model and nominal banks unchanged for the independent evaluation.

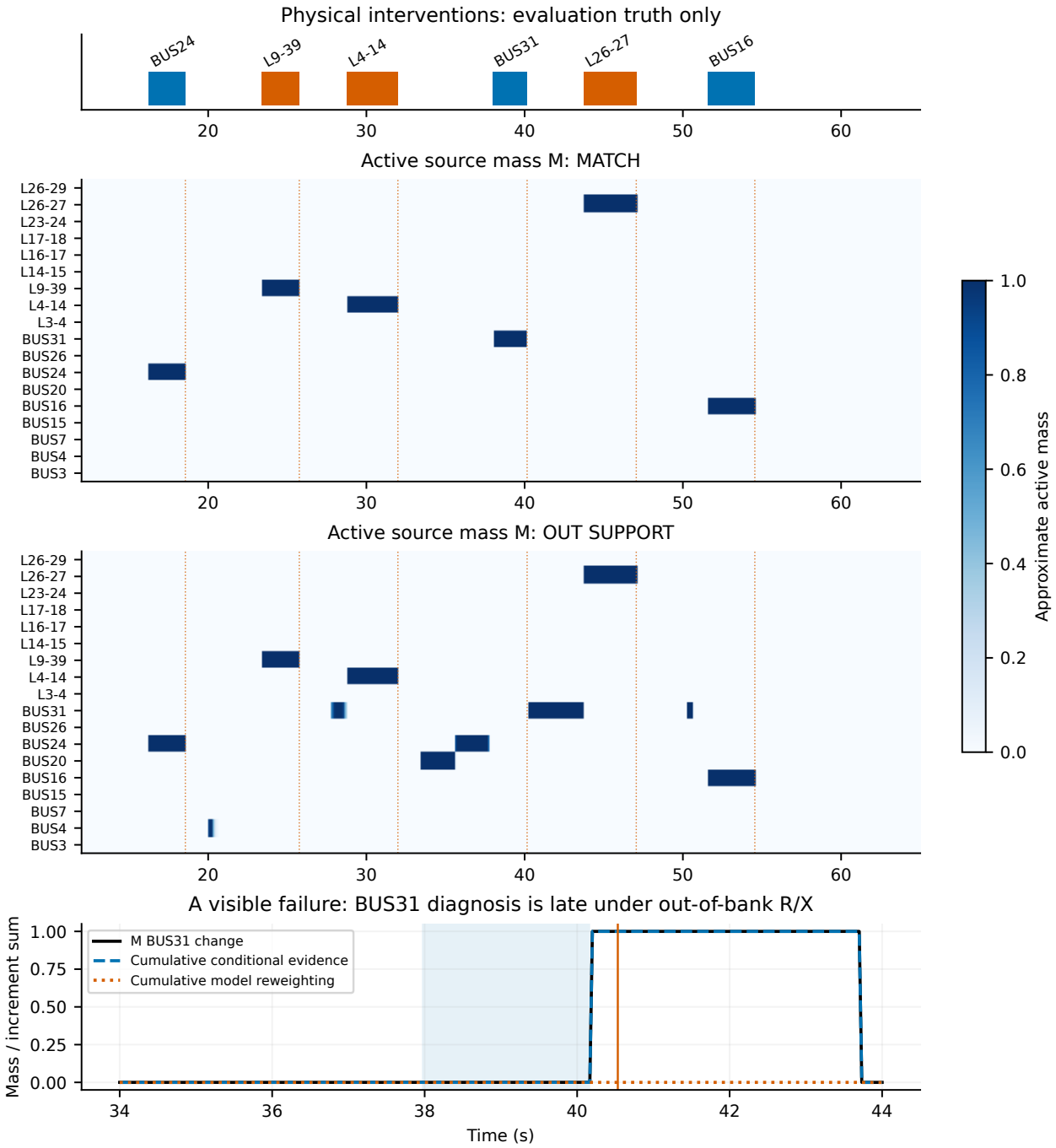


FIGURE 4. **From a continuous sequence to source attribution. Top: reserved physical ON–OFF intervals in one development parent. Middle: active masses for nominal-network observations (MATCH) and mixed R/X changes outside the bank (OUT SUPPORT). Orange dotted lines mark true OFF for evaluation only. Bottom: BUS31 in the latter record; the cumulative terms of $\acute{M}$, rebased at 34 s, sum to the change in active mass. Diagnosis arrives after OFF. The figure shows an eventual correct source that fails timely attribution; the color scale represents approximate mass, not calibrated probability.**

APPENDIX M DETECTABILITY COMPUTATION AND NOISE-STRESS PROTOCOL

a: Operators.

The information analysis of the main article uses the nominal bank built by the evaluated implementation for one held-out calibration record of a later campaign whose runtime is byte-identical to the independent evaluation runtime. The independent-evaluation banks had been evicted by the storage-retention policy; the nominal operators differ between records only through the prefix-estimated operating factors, which remain within 2×10^{-5} of unity.

b: Signatures.

For each source i, the noise-free switched predictor is propagated from a zero pre-event deviation with the ON operator at half-phase 0 followed by the stay operator (loads at severity 0.1). Each candidate filter $j \in \{0, \dots, 18\}$ is run on these reports with the quiet-regime gains K_o and inverse innovation covariances S_o^{-1} of the implementation. The cumulative sums $\sum e^\top S^{-1} e$, $\sum s^\top S^{-1} e$ and $\sum s^\top S^{-1} s$ give the signature information of the main article with the competitor's severity profiled on $[-0.2, 0.2]$. As a check, the true-history filter at the true severity leaves a residual energy below 10^{-8}. The idealized score reference uses the birth hazard $1 - \exp[-(1/60)/30]$ shared equally by the 18 cells.

c: Reset cost.

For every intervention of the controlled study, the noise-free truth is propagated through ON, the active interval, OFF and the gap until the next onset. At the inferred closure time of the carry arm, the first 114 coordinates of the true augmented state are zeroed, and the no-intervention and all birth filters are run on the remaining truth in the recovery covariance regime. The largest profiled gain of a birth over the no-intervention history is compared with the idealized score reference $\lambda^\star$, not a publication guarantee. Harm in the reset arm is an unmatched alarm between the OFF of the intervention and the next onset, or a lost timely attribution of the next intervention.

d: Noise stress.

Observations are regenerated from the stored deterministic trajectories with the simulator's own noise stream and seed: $\varepsilon = c + \alpha(v - c)$ for multiplier α, where v is the stream and c its normalization center, exactly as the campaign exporter. The $\alpha = 1$ observations reproduce the original files to 6×10^{-10} and the inference reproduces all 18 published messages of the parity record. The independent-noise scenarios add zero-mean Gaussian perturbations per PMU and report, with relative magnitude and angle (radian) standard deviations of 5×10^{-4} or 10^{-3}; the added voltage-angle noise is passed through the simulator's causal 21-report quadratic frequency and ROCOF estimator. For a reference phasor z, this perturbation gives $\widetilde{z}/z = (1 + \epsilon_m)e^{\mathrm{i}\epsilon_\theta}$, with independent $\epsilon_m, \epsilon_\theta \sim \mathcal{N}(0, \sigma^2)$. Consequently, the RMS TVE of the added perturbation is

$$\sqrt{\mathbb{E}|\widetilde{z}/z - 1|^2} = \sqrt{\sigma^2 + 2(1 - e^{-\sigma^2/2})} \simeq \sqrt{2}\,\sigma.$$

The two settings therefore add approximately 0.071% and 0.141% RMS TVE; these are neither per-report bounds nor the total error including the original noise. The nominal bank is rebuilt from the stored derivative matrices with the noise scale estimated from the noisy prefix. The operating point remains at its $\alpha = 1$ estimate because derivatives exist only there. Estimated angle noise scales of these scenarios are about 48 times the calibrated scale, outside the implemented support $[0.1, 10]$; for this test only the support was widened to $[0.1, 1000]$. Two jobs ran concurrently with another campaign on the same desktop CPU, so the reported processing rates are conservative.

e: Reproduction.

Every number and figure of the analysis is regenerated from the stored operators and outcomes by the reproducibility package, which also contains the noise-stress generator.

TABLE 17. Noise-stress outcomes with fixed publication thresholds and widened startup noise-scale support. σ is the standard deviation of each added relative-magnitude and radian-angle component.

Scenario	Load	Line	Extras	Normal
$\times 1$	12/12	12/12	0	0
$\times 5$	11/12	12/12	0	0
$\times 10$	11/12	12/12	0	0
$\times 20$	12/12	12/12	0	0
$\sigma = 0.0005$	3/12	12/12	0	0
$\sigma = 0.001$	0/12	10/12	0	0

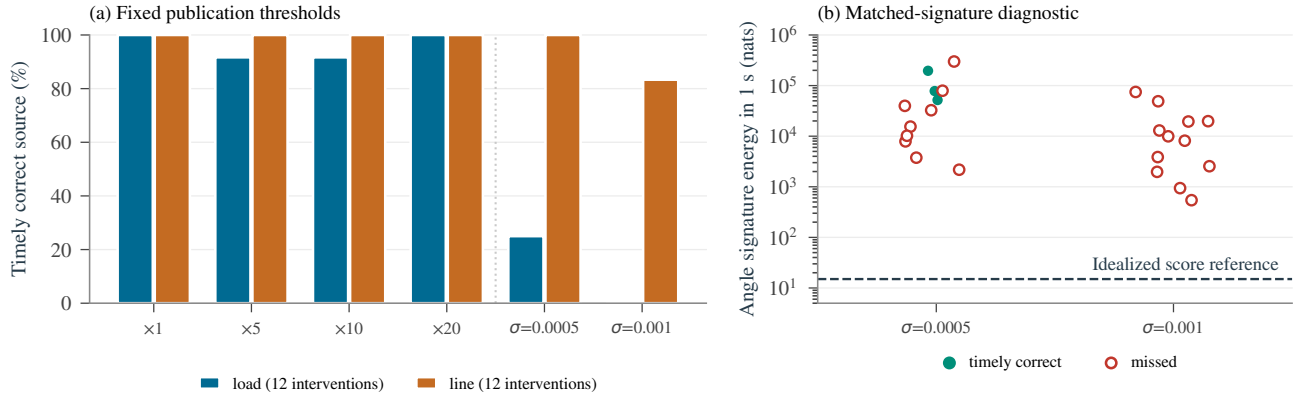


FIGURE 5. Nominal bank under increased PMU noise, with fixed publication rules and widened startup noise-scale support. (A) Timely attribution under uniform noise scaling and added independent noise, with component standard deviation σ (relative magnitude and radian angle). (B) One-second matched-signature energy using only the added independent angle noise, against the observed outcome. The dashed line is an idealized score reference, not the bank's publication threshold.

APPENDIX N WHAT TERMINAL CURRENTS ADD TO VOLTAGE SIGNATURES

This diagnostic compares the observation sets available to distinguish fixed source responses. It uses the nominal switched predictor and its implemented terminal map, including the provisional bus-5 assignment. Loads have a fixed 10% change; lines use their opening operators without an adjustable amplitude. The onset and pre-event state are supplied. These are model-based separations, not attribution outcomes of the sequential detector.

a: A common temporal noise reference.

To compare nested observation sets, all source means share one reference covariance. The quiet no-intervention predictor defines

$$r_k = J_0 m_k + e_k, \tag{25}$$

$$m_{k+1} = T_0 m_k + K_0 e_k, \qquad e_k \sim \mathcal{N}(0, S_0), \tag{26}$$

with independent innovations and fixed initial predictive state. Here m_k is the predictor's conditional mean, rather than a new physical plant state. Stacking reports gives $\Sigma = L(I \otimes S_0)L^\top$, where L has identity diagonal blocks and $L_{kr} = J_0 T_0^{k-r-1} K_0$ for $k > r$. Thus selecting voltage and current coordinates retains the induced cross-time covariance. It is not equivalent to selecting coordinates from innovations produced by a filter that already used all channels.

b: Conditional separation.

Let d be the difference between two fixed mean trajectories. Partition the selected report coordinates into voltage V and current I and set

$$S_{I|V} = \Sigma_{II} - \Sigma_{IV}\Sigma_{VV}^{-1}\Sigma_{VI}, \tag{27}$$

$$d_{I|V} = d_I - \Sigma_{IV}\Sigma_{VV}^{-1} d_V. \tag{28}$$

For a positive-definite common covariance,

$$Q_{VI} = Q_V + Q_{I|V}, \tag{29}$$

$$Q_V = \tfrac{1}{2} d_V^\top \Sigma_{VV}^{-1} d_V, \tag{30}$$

$$Q_{I|V} = \tfrac{1}{2} d_{I|V}^\top S_{I|V}^{-1} d_{I|V}. \tag{31}$$

The second term is nonnegative and measures the contribution of current coordinates after conditioning on voltage for this pair. It is a standard Gaussian identity. It does not establish the contribution of currents after separately fitting severity or other nuisance parameters, and it does not decompose the actual bank's model-dependent likelihoods. The nearest competing source may also change with the observation set.

c: Executed calculation.

The operators and quiet covariance come from one nominal calibration bank built with a 300-report PMU prefix. No test outcomes were used. We evaluated all 153 unordered source pairs at horizons of 3, 15 and 30 reports (0.1, 0.5 and 1 s at 30 reports/s); the stored table includes both directions. These are deterministic comparisons, not independent event replicates. At 1 s the ratio of minimum pairwise separation with V+I to that with V has median 1.35 and range 1.03–8.41 across sources. For each source's nearest V+I competitor, the conditional current term accounts for a median 18.0% of joint separation, with range 2.6%–76.6%. The two summaries answer different questions: the first permits the nearest competitor to change; the second holds the compared pair fixed.

The implementation reconstructs the raw-report covariance recursively and checks it against an explicit innovation transformation. Their relative discrepancy is below 6×10^{-16} at three reports. Full-channel quadratic forms agree with the null-innovation calculation to 3.1×10^{-15} relative error over all pairs and horizons; the conditional decomposition agrees to 1.4×10^{-11}. All nested-channel distances are nondecreasing. The source package contains the compact operator snapshot, channel maps, checksums, calculation and figure-generation scripts.

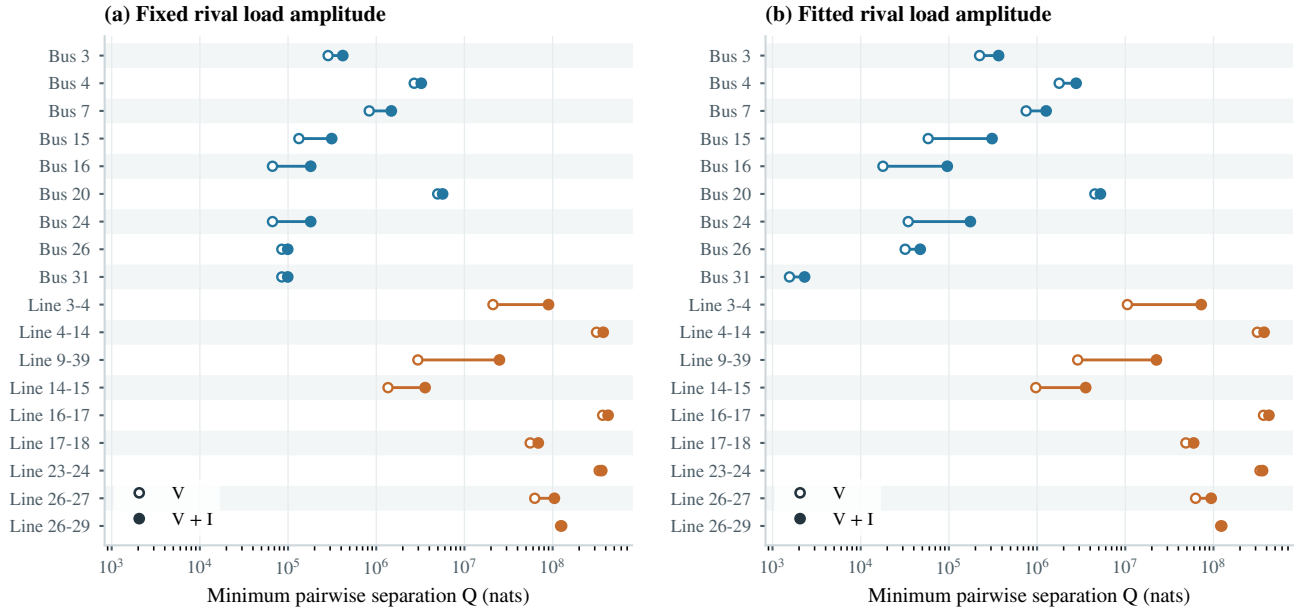


FIGURE 6. **Terminal-current information under a common nominal covariance at 1 s. Blue denotes load changes and orange line openings. (a) Minimum fixed-amplitude pairwise separation using voltage alone (open circles) or voltage and current (filled circles); the minimizing competitor may differ. (b) Rival load amplitudes are fitted over $[-0.2, 0.2]$, while line openings retain their fixed operators. Onset, initial state and true source amplitudes are fixed in both panels; these values do not measure detector performance.**

d: Allowing a rival to fit severity.

For a load rival with mean $\mu_j(b) = o_j + s_j b$, set $d = \mu_i - o_j$. Under the same common covariance, its best bounded fit is

$$b^* = \mathrm{clip}_{[-0.2,0.2]} \frac{s_j^\top \Sigma^{-1} d}{s_j^\top \Sigma^{-1} s_j}, \qquad s_j \neq 0. \tag{32}$$

The profiled separation is $\widetilde{Q}_{i|j} = \frac{1}{2}\|d - s_j b^*\|^2_{\Sigma^{-1}}$. Line rivals retain their complete opening trajectories, with no fitted amplitude. This is deterministic profiling, not integration of the truncated severity prior used by the detector.

For each common feasible amplitude b, the conditional decomposition gives $Q_{VI}(b) = Q_V(b) + Q_{I|V}(b) \geq Q_V(b)$. Hence $\min_b Q_{VI}(b) \geq \min_b Q_V(b)$. This does not imply that the two separately minimized terms add to the joint minimum: both must use the same amplitude. The inequality also holds after minimizing over the same rival catalog, although the nearest rival may change.

We repeated all source comparisons with bounded rival load severity, keeping true load changes at 10% and the same covariance and horizons. Every analytic load fit agreed with an independent scalar numerical optimizer to relative error below 10^{-12}. At 1 s, the ratio of minimum V+I to V separation has median 1.53 and range 1.03–7.82. With V+I, 4 sources change their nearest rival after fitting severity. The bus-31 source retains only 2.3% of its fixed-amplitude minimum separation: its nearest rival changes from bus 26 to bus 7. Thus a large distance between fixed responses can conceal ambiguity between source and magnitude. These nominal comparisons do not establish how often that ambiguity causes an attribution error.

e: Interpretation.

The reference covariance is fixed across sources; candidate-specific covariance terms, uncertain onset, severity marginalization, pruning, publication thresholds and false alarms are outside this diagnostic. The full set contains 16 voltage, 15 current and 16 frequency/ROCOF coordinates. Those derivative coordinates follow the implemented measurement model; their incremental separation is not a statement about independent physical sensors. An empirical channel ablation would require rebuilding each filter with its own observed coordinates and calibrating its publication rule before a new independent test.

A. RECOVERY UNCERTAINTY AND APPARENT SOURCE INFORMATION

A local Gaussian model explains why preserving a recovery mean alone can overstate confidence in a new source. Suppose, conditionally on an earlier history,

$$r = Ox + sa + \epsilon, \qquad x \sim \mathcal{N}(0, P), \quad \epsilon \sim \mathcal{N}(0, R),$$

where x and ϵ are independent and P, R are positive definite. Here R excludes the initial-state uncertainty represented by P. Marginalizing that state gives $\Sigma(P) = R + OPO^\top$. For a fixed local source direction s, Woodbury's identity gives

$$\begin{aligned}\mathcal{J}(P) &= s^\top \Sigma(P)^{-1} s \\ &= s^\top R^{-1} s - b^\top (P^{-1} + O^\top R^{-1} O)^{-1} b.\end{aligned} \tag{33}$$

Here $b = O^\top R^{-1} s$. The nonnegative subtracted term measures how much of the apparent source information can be explained by uncertain recovery coordinates. If $P_2 \succeq P_1$, then

$\mathcal{J}(P_2) \leq \mathcal{J}(P_1)$ for this fixed direction and noise law. This standard Gaussian identity explains a possible confidence error; it does not prove that a larger covariance improves attribution. Likelihood normalization, competing sources and false alarms still matter.

This decomposition is a separate explanatory model. The preceding channel diagnostic already uses a covariance induced by the complete null innovation model. Adding the bank's state covariance to it again would count the same predictive uncertainty twice. Likewise, an uncertainty already marginalized in Σ must not also be fitted as a free recovery direction without redefining the statistical model. The implemented state–severity correlations are specified in Section A-C; independence is not assumed for those stored posterior variables.

Port-Hamiltonian formulations provide a physical interpretation of network interconnection and energy balance [2]. Their reduced swing-model energy is not automatically an energy function for the present lossy DAE with exciters and governors. Similarly, the input-change versus topology-change distinction is exact only within its stated model: a finite impedance-load change can alter algebraic admittance and the local linearization. The implemented load direction is an affine approximation of that change. Energy residuals and reconstructed hidden voltages are therefore not appended as independent likelihood factors to the PMU reports that produced them.

APPENDIX O MEASUREMENT-INTEGRITY DIAGNOSIS

a: Statistics.

For each PMU and report, the three phase phasors of voltage and of current are transformed to symmetrical components (X_0, X_1, X_2). For each voltage or current set, the parity statistic is $q = 2(|X_2|^2+|X_0|^2)/(\overline{|X_2|^2 + |X_0|^2})$, where the overbar is the mean over the first 90 reports (3 s) of the record. With known scale and independent Gaussian per-phase noise, the normalized statistic has an idealized $\mathrm{Gamma}(2, 1)$ tail, $P(q > x) = (1 + x)e^{-x}$. The scale is estimated from the prefix, however, and the test uses the maximum of the voltage and current statistics. Thus the prespecified threshold $x = 23.9$ is a nominal 10^{-9} single-statistic tail reference, not an exact report-level false-alarm probability. Under the ideal marginal model, a union bound gives at most 2×10^{-9} per PMU-report for the maximum. The variance-inflated, post hoc operating point doubles the threshold to 47.9. A report is flagged if either statistic exceeds the threshold. The corrupted phase is the phase whose phasor has the largest projection on X_0.

b: Network-coupling bound.

The admittance matrix of the 39-bus network, including line charging and transformer taps, is augmented with the base-case loads as constant impedances and the generator transient reactances of the New England system on 100 MVA [3]. The bound c_p of the main article is the smallest, over the fault buses at which PMU p is the most affected PMU, of the largest sag ratio at another PMU. Table 18 lists the values. For the three-phase fault at bus 39 of the competition record, the model predicts sag ratios of 0.10–0.29 at the other PMUs. At the deepest fault report the observed ratios are 0.12–0.43, so the bound is conservative there. At the first fault report the largest ratio is only 0.11, 0.37 times c_{39}, because the phasor estimates of the distant PMUs lag the faulted bus. This motivates the margin of one quarter. A balanced change is tested when it exceeds 10% of the PMU's startup level relative to the median of the previous 30 reports. It is an integrity event when the largest change elsewhere is below $c_p/4$ times that change. While a coupling-consistent change is present at any PMU, and for 2 s afterward, sequence parity is suspended.

TABLE 18. Network-coupling bound c_p, the fault bus that attains it and the PMU most coupled to it.

PMU bus	c_p	Fault bus	Most coupled PMU
2	0.39	30	29
5	0.93	5	6
6	0.94	6	5
10	0.71	32	6
19	0.32	34	22
22	0.34	22	19
29	0.16	29	2
39	0.29	39	5

c: Unlabeled incidents.

At the variance-inflated operating point all detections outside labeled episodes are listed here. Single-report current-magnitude spikes occur in phase B of PMU 2 at 11.57 s (2,485 A) and 12.33 s (53 A), against a median of about 585 A, and in phase A of PMU 39 at 5,378.77 s (702 A) and 5,379.03 s (710 A), against 245 A. A balanced 12% jump appears at PMU 29 in the frame immediately before the dropout at 536.57 s. From 2,776.90 to 2,779.10 s, after the reconnection that ends at 2,776.07 s, the three voltage magnitudes of PMU 29 collapse to zero and include negative values. The other PMUs change by less than 0.7% during this interval. Two single-report flags coincide with switching instants: PMU 22 0.47 s after the line opening, and PMU 2 at the onset of the generation change at bus 2.

d: Simulated integrity stress.

The four held-out event records and two normal records of the noise-stress study were used with the calibrated noise ($\times 1$), whose outputs reproduce the stored campaign predictions. Corruptions rotate over the eight PMUs: three in the quiet tail of each event record (75, 90 and 105 s), one 1 s after the third intervention onset, and twelve in each normal record (every 50 s from 30 s). Dropouts rotate over PMUs 29, 5, 19 and 39 in the event records, and over PMUs 6, 10, 22 and 29 in the normal records, each lasting 27 s. The bank, its startup estimates and its operators are those of the stored campaign; the 300-report startup is unchanged in every scenario.

A. INDEPENDENT-EVALUATION FIGURES

B. MEASUREMENT INTEGRITY

The competition data also contain integrity events: frames that never arrive and frames whose values are corrupted. Write the received report of PMU p as

$$y_{p,k} = M_{p,k}\big(h_p(x_k) + v_{p,k} + \beta_{p,k}\big), \tag{34}$$

where $M_{p,k} \in \{0,1\}$ is availability and $\beta_{p,k}$ a corruption that does not enter the network equations. Availability is observed through the frame flag, so a missing PMU is identified exactly. What it removes is information, and the same quantity governs corruption.

Proposition 1 (Integrity isolation): Let the physical-source and no-intervention hypotheses have the same Gaussian covariance, block diagonal across PMUs. Let corruption of PMU p be an unrestricted additive mean shift over the window. The minimum Gaussian Kullback–Leibler divergence from physical source i to the no-intervention hypothesis with PMU p corrupted is $\Lambda_{i0}^{(-p)}$, the mean-separation information for i computed without PMU p's channels, using the same fixed covariance and severity profile. This is also the information for i when PMU p is missing.

Proof. With common covariance, the covariance terms in the Gaussian KL divergence cancel, leaving one half of the covariance-weighted squared mean difference. Minimizing over an unrestricted β_p removes the component supported on PMU p. For block-diagonal covariance, the residual quadratic form is exactly the mean-separation information in the remaining PMUs. □

For a common joint Gaussian covariance Σ and a free shift on the corrupted coordinates, minimizing the quadratic separation over that shift gives $\frac{1}{2} r_c^\top \Sigma_{cc}^{-1} r_c$, where c denotes the remaining coordinates. This uses the marginal covariance of the clean observations; equivalently, Σ_{cc}^{-1} is the Schur complement of the corrupted block in the joint *precision* matrix. Conditioning on corrupted values as though they were clean would answer a different question. This identity does not provide a general integrity detector: it states what information remains under a free additive corruption at one PMU. The two monitors below use narrower physical checks. *Sequence parity.* For balanced interventions (load, line, generation, three-phase fault) the balanced network produces no negative- or zero-sequence voltage or current. These sequence components are therefore ideal-model parity relations in the sense of analytical redundancy [4]: they vanish in this balanced observation model and sensitive to corruption of a single phase. A step ϵ in one phase magnitude gives $|X_2| = |X_0| = \epsilon|X|/3$. The corrupted phase is the one whose phasor is most aligned with X_0. With known scale and independent Gaussian per-phase noise, $q_k = (|X_2|^2 + |X_0|^2)/\sigma^2$ is $\mathrm{Gamma}(2,1)$-distributed. In the implementation the scale is estimated from a 90-report prefix and the statistic is the maximum of voltage and current values, so the resulting threshold is only a nominal single-statistic tail reference, not an exact report-level false-alarm calibration. In the simulated causal cohorts, phases B and C are reconstructed from phase A under the balanced-waveform assumption; they do not provide independent sequence observations. The sequence-parity evaluation below therefore applies only to the separately exported three-phase competition record. *Network coupling.* A modeled single bolted fault that changes the positive-sequence voltage of PMU p by Δ_p changes the other PMUs in proportion to the network impedance matrix $Z = Y^{-1}$. Here Y includes constant-impedance loads and generator transient reactances [3], and a bolted fault at bus f gives $\Delta_j/\Delta_p = |Z_{jf}/Z_{pf}|$. Hence

$$\max_{j\neq p} \Delta_j \;\geq\; c_p\, \Delta_p, \qquad c_p = \min_{f\in\mathcal{F}_p} \max_{j\neq p} \left|\frac{Z_{jf}}{Z_{pf}}\right|, \tag{35}$$

with $\mathcal{F}_p$ the buses at which p is the most affected PMU. For the IEEE 39-bus network c_p ranges from 0.16 (PMU 29) to 0.94. A balanced excursion that violates (35) by a wide margin is inconsistent with the modeled set of single bolted faults. It does not rule out other physical disturbances or report-window effects. Both relations are classical bad-data reasoning [5], applied here to report-rate phasors without a state estimator.

TABLE 19. Labeled integrity episodes of the competition record and their diagnosis. Delay is in reports after the first labeled report; "flagged" counts diagnosed reports within the episode; phase is identified (labeled-data phase). Every episode was located at the correct PMU.

Type	PMU	Start (s)	Length (s)	Delay	Flagged	Phase
Corrupted	2	3.3	0.73	0	22/22	C (C)
Corrupted	39	334.4	0.77	0	23/23	A (A)
Corrupted	39	336.5	1.07	0	15/32	A (A)
Missing	29	536.6	27.07	0	812/812	–
Missing	29	614.3	29.10	0	873/873	–
Missing	29	2584.7	191.43	0	5743/5743	–
Missing	29	2935.8	59.07	0	1772/1772	–

C. MEASUREMENT INTEGRITY

Competition record. The competition record contains 90 min of reports from the eight PMUs (161,379 reports per PMU). Its labels mark four dropout events of PMU 29 (9200 reports) and three corrupted-data episodes at PMUs 2 and 39. A separate class 6 label composes a missing PMU with the generation event at bus 2 and is counted as an additional event by the competition scorer; the integrity monitor still sees the same missing frames. We applied the two parity relations of Section O-B and the frame flag. Separate voltage and current noise scales for each PMU were estimated from the first 3 s of the record; there was no training and labels were used only for scoring. Table 19 and Fig. 10 give the results. All 4 dropouts and all 3 corruptions were diagnosed at their first report and located at the correct PMU. The corrupted phase was identified in 3/3 episodes. The third corruption is intermittent, with a median step below 1%, and is flagged in 15 of its 32 reports.

The three-phase fault at bus 39 produces negative-sequence transients at seven PMUs at once. Its four reports with a

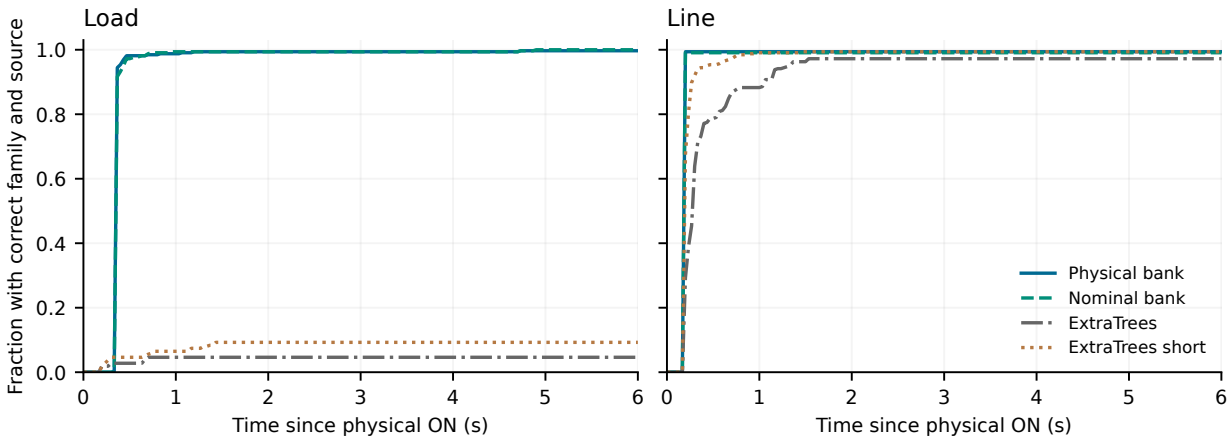


FIGURE 7. **Distribution of correct diagnosis time, using all 324 replays per true family in each denominator. An incorrect or absent diagnosis contributes no mass, and late correct decisions are included, so the curves differ from the before-OFF endpoint.**

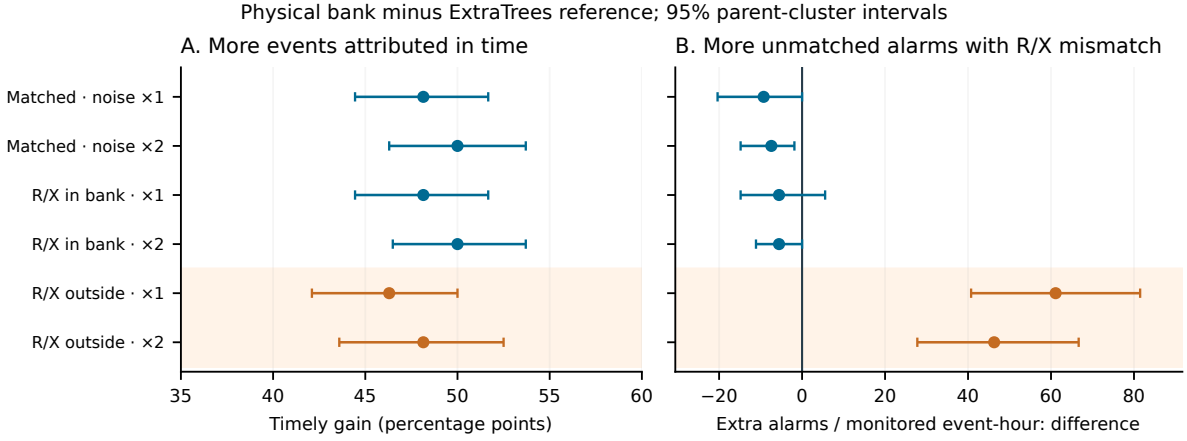


FIGURE 8. **Physical bank minus the ExtraTrees reference in the six paired strata. Points and 95% intervals come from 10,000 physical-parent resamples. Left: percentage-point difference in timely attribution; right: difference in unmatched alarms per monitored event-record hour (the 12-s startup is excluded). Orange rows mark out-of-bank R/X conditions.**

balanced change above 10% lie between 0.37 and 1.5 times the coupling bound, above the quarter of the bound used by the test (Fig. 10B). The lowest ratio occurs at the first report, where the phasor estimation window has reached the faulted bus before the others. The fault is therefore recognized as physical, and parity is suspended while it lasts. The threshold $q > 23.9$ is the nominal 10^{-9} tail point for one idealized sequence statistic. The implemented report statistic is the maximum of voltage and current, so its ideal marginal tail is at most 2×10^{-9} before accounting for scale estimation. Across the eight PMUs this record contains about 1.29 million monitored PMU-reports, for an ideal-model expectation below 0.0026 exceedances; we observe 19 isolated single-report values with $23.9 < q < 47.9$ (about 7.4×10^3 times that reference expectation). This shows that the nominal Gaussian tail model does not calibrate this record. Doubling the assumed noise variance ($q > 47.9$) removes these exceedances and keeps every labeled episode. This operating point was chosen after inspection and is not an independently calibrated FAR. Two single-report flags remain at switching instants: the line opening, and the generation change at the bus of PMU 2. This gives 0.022 false alarms per minute.

The monitor also flags 6 incidents that the labels leave as normal operation. Four are single-report current spikes: 2,485 A and 53 A in phase B of PMU 2, against about 585 A, and 702 A and 710 A in phase A of PMU 39, against 245 A. One is a corrupted frame just before a dropout of PMU 29. The last is a 2.2-s interval after a reconnection in which all three voltage magnitudes of PMU 29 fall to zero, some of them negative, while every other PMU changes by less than 1%. A bolted fault at bus 29 would change another PMU by at least $c_{29} = 0.16$ times the change at PMU 29. The excursions are inconsistent with the tested single-fault model. We report them as unlabelled integrity candidates, separately from false alarms.

Physical bank under integrity events. The physical bank cannot run on the competition record. Its startup test rejects the first 10 s of the record: the ROCOF channels of two PMUs are constant or quantized, and the angle reference drifts by 0.31 deg/s with respect to the calibrated model.

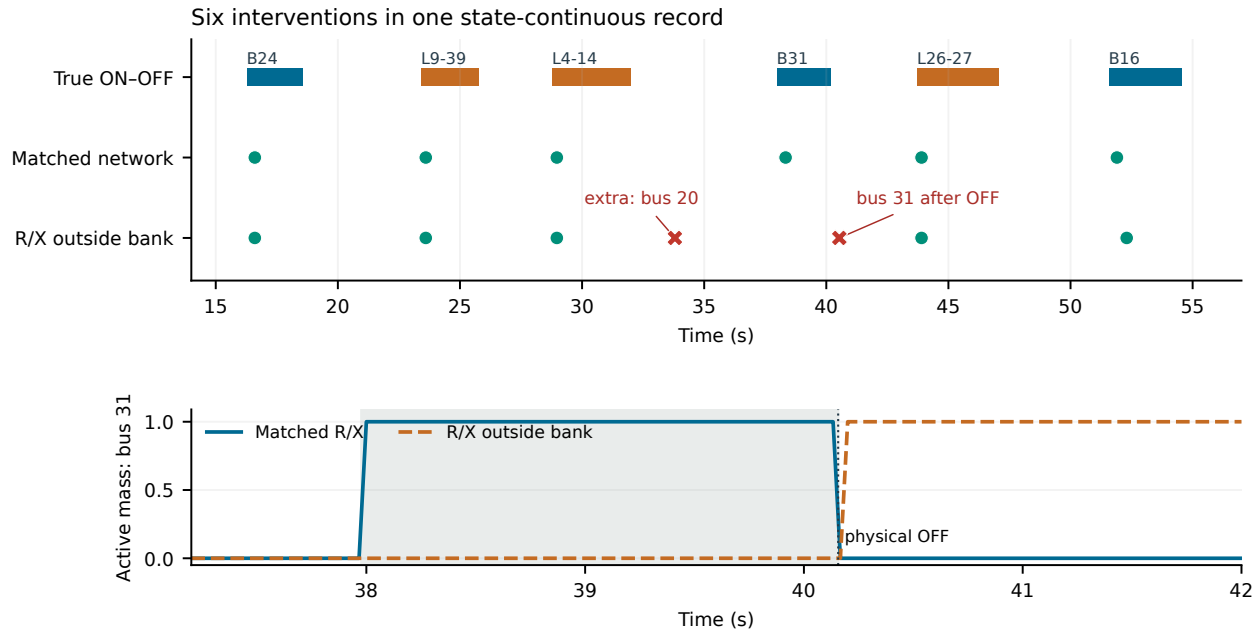


FIGURE 9. **One continuous development trajectory, shown for interpretation. Top: physical ON–OFF intervals and emitted diagnoses under matched and out-of-bank R/X. Bottom: bus-31 active mass with the true intervention shaded. A diagnosis after OFF misses the timely endpoint even when it identifies the source. True timing enters only this retrospective display.**

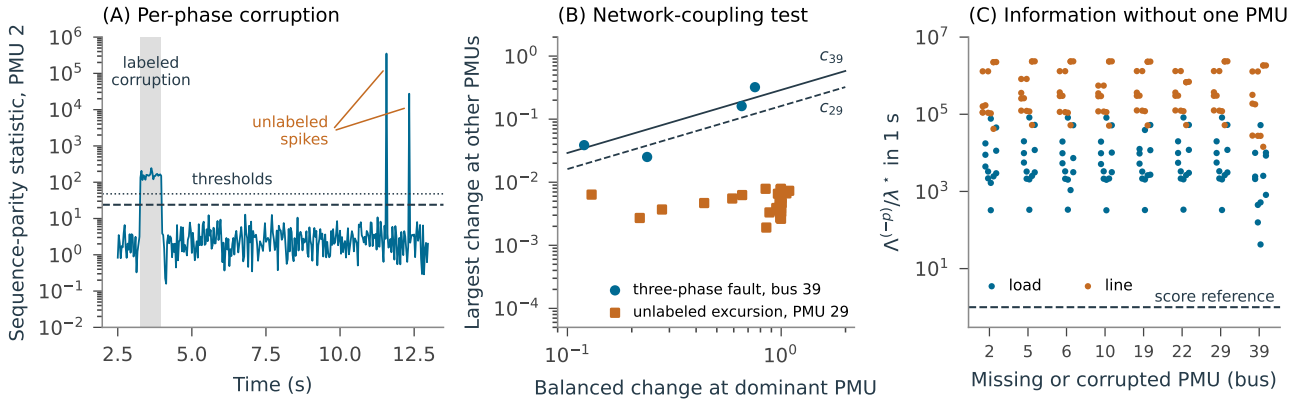


FIGURE 10. **Measurement integrity from network physics. (A) Sequence-parity statistic of PMU 2 on the competition record: a labeled single-phase corruption and two unlabeled single-report current spikes; dashed and dotted lines are the prespecified and variance-inflated thresholds. (B) Balanced voltage changes larger than 10%: reports of the three-phase fault at bus 39 lie near the network-coupling bound c_{39} of (35) and above the quarter of it used by the test, whereas the unlabeled collapse at PMU 29 lies two orders of magnitude below c_{29}. (C) Information for each held-out simulated intervention when one PMU is missing or corrupted (Proposition 1), relative to the idealized score reference $\lambda^\star$, under independent component noise with standard deviation 0.0005 (relative magnitudes and radian angles).**

We therefore inserted the competition's integrity events into the four held-out simulated event records and two normal records of the noise-stress study, and ran the bank with its code unchanged. *Corruption*: we added 40 0.75-s steps of $+2.4\%$ or -2.6% to the positive-sequence voltage magnitude of one PMU, the positive-sequence equivalent of the two labeled corruptions. They rotate over all PMUs, and four fall inside interventions. None of them produced a physical alarm, and the timely attributions were unchanged (12/12 loads, 12/12 lines). *Missing data*: one PMU was removed for 27 s, starting 2 s before an intervention, and replaced by its last valid report, the imputation suggested by the competition guide. Of the 13 interventions that began while a PMU was missing, the bank attributed 1 in time, and the dropouts caused 10 unmatched alarms on the event records. Attribution did not recover after the PMU returned: 0 of the 7 later interventions in these records were attributed in time, against 4 of 4 before the dropout. On the completed normal record the two held intervals produced 2 alarms (the second normal record of this scenario was not run). A held value is a measurement that contradicts the network response, not an absent one; the bank therefore needs the deletion of Proposition 1, an availability-switched observation operator, rather than imputation.

Together these results separate the integrity mechanisms

from the physical ones. Missing data are identified exactly and leave ample nominal signature separation, but the implemented bank cannot use it when fed held values; it needs the availability-switched operator. Corruption is flagged by departures from the assumed measurement and network model. It did not mislead the bank in simulation, and on the competition record it revealed incidents that the labels miss.

D. ALL EVENT TYPES: NETWORK PHYSICS WITHOUT TRAINING

Because the bank cannot start on the competition record (Section O-C), we built a second attributor from the provided network data alone, with no trained parameter, for every event type of the guide. The network is the admittance matrix of the provided power-flow case, with loads as constant impedances and generators as transient reactances behind constant voltages [3]. The integrity layer runs first; unhealthy PMUs are deleted from the fits (Proposition 1), not imputed. The physical layer uses the two time scales of the network equations. *Algebraic jump:* the one-report jump of the PMU voltages and branch currents is fitted by an injection at each of the 39 buses or the removal of each of the 46 branches, removed with its full pre-event current at both terminals; a sag below 0.8 pu is a fault, located by the profile $|Z_{jf}/Z_{ff}|$. *Quasi-steady state:* an episode that does not return within 15 s is fitted 15–25 s after onset by the linearized AC power flow, with generators sharing the imbalance in proportion to dispatch. This separates load changes, generation changes and generator trips, under plausibility limits (a load cannot change by more than its base value, nor a generator by more than its dispatch). A reversed jump, or a return to the pre-episode state, closes an episode. Attributor I detects episodes from the change of the angle differences. Attributor II detects one-report jumps after subtracting the local trend of the neighboring report changes, which separates switching from the smooth electromechanical response. Decisions are taken 0.5 s (switching) or 25 s (persistent) after onset and written back to it, as in the guide's offline scoring; each record takes seconds.

Protocol. No label was used to fit a parameter, but the competition record was not blind. Attributor I took 6 logged iterations on it. Attributor II was designed on the training and calibration records of a simulated cohort and checked on the competition record in two further iterations. Every iteration is stored with its code hash and score; the competition record is thus a validation set. The confirmatory test is the stored test split of that cohort, generated before either attributor existed. Each attributor was fixed and run once, and its predictions were hashed before the schedule was read.

Results. On the competition record (Table 20) both attributors detect, classify and locate all 12 guide-defined event labels of that one record: three corruptions, four dropouts, one three-phase fault, one line outage, one generation change, one load change, and one simultaneous missing-PMU/generation event (class 6). The composite label is counted separately from the coincident generation label, as in the competition scorer. The submitted system located 7 of them. For the generator case, the guide's event label is associated with bus 2. The attributor fits a bus-level effective injection at bus 2 (scale 1.022, with no separate amplitude fit); this is a guide-level Top-1 localization and does not identify a specific generator unit. The load at bus 7 is separated from generation by its reactive-to-active ratio. Six of the eight false-alarm episodes of Attributor II are the impossible measurements of Section O-C, a seventh follows the collapse at PMU 29, and the eighth is an unlabeled change at 3580 s seen at every PMU. Labeled boundaries that extend beyond the observable change keep macro-F1 at 0.88.

TABLE 20. All event types on the competition record (12 guide-defined labels: one fault, one line outage, one generation change, one load change, four dropouts, three corruptions and one simultaneous missing-PMU/generation event). The composite label is scored separately from the coincident generation label. Report-level scores follow the competition guide; false alarms are episodes outside every label.

Method	Class	Top-1	F1	FA/min	Macro-F1
Submitted system	11/12	7/12	0.192	0.18	0.21
Attributor I	12/12	12/12	0.981	0.10	0.90
Attributor II	12/12	12/12	0.979	0.10	0.88

TABLE 21. Stored simulated cohort, evaluated once: 72 load and line pulses (2–5 s, six per record) in 12 records, and 12 normal records (2 h). Top-1 counts events with the right family and exact source. The dynamic bank uses the plant model and an 18-source catalog; the open search ranks all 39 buses and 46 branches.

Method	Detected	Top-1	Lines Top-1	Loads detected	Loads Top-1	Normal alarms
Attributor I, open	13/72	0/72	0/36	4/36	0/36	0
Attributor II, open	48/72	28/72	24/36	12/36	4/36	0
Attributor II, catalog	48/72	40/72	32/36	12/36	8/36	0
Dynamic bank, catalog	72/72	72/72	36/36	36/36	36/36	0

The stored cohort has 72 load and line pulses of 2–5 s, six per record with gaps of 3–8 s (Table 21). Attributor I fails there: its quasi-steady window opens after the pulses end. Attributor II detects all 36 line pulses and locates 24 in an open search over 85 candidates, or 32 within the bank's 18-source catalog. Every remaining line error involves one of three branches whose outage, seen from the eight PMUs, is nearly collinear with an injection at an adjacent bus. It detects only 12 of the 36 load pulses, because a change of 3–14% of one load moves the one-report phasors by less than the startup noise. Longer windows would recover them, but there the response is no longer algebraic. The dynamic bank models that response and attributes all 72 pulses, at 9.2 h of computation against 4 s. No method alarmed in the normal records.

Network physics alone thus attributes every event type that is large enough for its algebraic jump or long enough to settle; short, small pulses need the dynamic model.

APPENDIX P OPERATING-POINT SIGNATURE EXPERIMENT

A. REFERENCE CORRECTION AND RESPONSE CORRECTION

The stream bank already builds local predictors at estimated operating points. A separate experiment tests why changing a reference alone may be insufficient. It compares three response approximations with identical observations, channels, timing, amplitude prior and inference horizon. It is a controlled test of a mixed derivative, not another version of the stream method.

Let η denote a scalar operating change and a the signed load factor. Central differences of complete nonlinear DAE trajectories give

$$D_i(\eta) \simeq \frac{y_i(\eta, h) - y_i(\eta, -h)}{2h},$$
$$E_i \simeq \frac{D_i(d) - D_i(-d)}{2d}. \tag{36}$$

The event follows the same ON/OFF schedule in every derivative trajectory. The fixed Q_i^0 is the coefficient of a^2, obtained from the centered second difference divided by $2h^2$. After removing the fixed common angle drift, the three predicted histories are

$$\mu_i^{\mathrm{N}}(a) = y_0 + aD_i^0 + a^2 Q_i^0,$$
$$\mu_i^{\mathrm{R}}(a) = \mu_i^{\mathrm{N}}(a) + \widehat{b},$$
$$\mu_i^{\mathrm{E}}(a) = \mu_i^{\mathrm{R}}(a) + a\widehat{\eta}E_i. \tag{37}$$

Both corrected variants use exactly the same prefix offset $\widehat{b}$. This comparator corrects the measured PMU reference; it is not the separate nonlinear AC hidden-state reconstruction algorithm in the historical study. Only the last changes the event signature. The first two consequently have identical incremental response models even when their absolute fits differ. Context $\widehat{\eta}$ is estimated by generalized least squares from the pre-event PMUs and the nominal context direction. No test context, source or severity enters inference.

B. TEST CASES AND DECISION CONTRACT

The research plant supplies all trajectories. The scalar η scales PQ operating injections and active PV dispatch; feasible initialization can change permitted operating setpoints, but not machine constants, line impedances, ZIP fractions or the PMU model. Central event/context steps are 0.005 and 0.01, with step-halving checks. Dictionary construction uses 57 trajectories. The test contains 85 physical cases: 80 load interventions and 5 controls at operating changes $-3\%, -2\%, 0, +2\%, +3\%$. Each load changes P/Q together by $\pm 0.1\%$ or $\pm 1\%$ at BUS3, BUS7, BUS16 or BUS24. None belongs to the prescribed voltage-measurement PMU set; the separate oriented current-terminal assignments remain fixed.

Each case is one ten-second integration, with ON at 3 s, OFF at 6 s and no state reset. The context fit ends at 2.5 s. Three fixed instrumental noise seeds produce 255 paired records. The nonlinear PMU frequency/ROCOF operator runs before its stored instrumental noise is added. Both corrected methods retain the same measured prefix, so an advantage cannot arise from supplying one method a better initial reference.

This experiment supplies the load family and ON/OFF times. It estimates source and continuous severity using the same bounded Gaussian prior and amplitude-integrated likelihood in all three variants. The no-event hypothesis is retained. The fixed colored-instrument transform is common; plug-in prefix estimates and clipped noise still limit probability calibration. Thus these decisions are not unknown-onset detector scores and normal-window decisions are not continuous false-alarm rates.

C. RESPONSE ERROR AND SOURCE DECISIONS

Relative to recentering alone, the mixed term reduces mean incremental response error: the paired difference is -1.15 percentage points (95% interval [-1.31, -1]). The corresponding joint-decision difference is 0 percentage points ([0, 0]), which does not establish a directional improvement in joint source decisions. The recentered and conditioned source decisions are already at the test ceiling. Median absolute relative severity error, evaluated at the generating source, changes from 1.2% to 0.329%. Its mean paired change is -0.738 percentage points ([-0.898, -0.582]). Both variants select the correct source in every tested event record, so this severity comparison does not discard source errors. Their source masses numerically saturate at one; zero recorded log-loss is not evidence of calibrated uncertainty.

The gain is not uniform across all comparisons: at the nominal operating point, estimating an unnecessary reference/context can worsen severity relative to keeping the correct fixed reference. Figure 12 retains that case rather than reporting only shifted operation.

The derivative stability checks change E_i by 0.283–0.427% when the context step is halved. Halving the event step in the two checked directions changes it by at most 0.00977%. These are local numerical checks, not a global smoothness or identifiability certificate. Percentages of event severity refer to the P/Q input factors; actual load power also responds to voltage under the fixed impedance law.

Paired intervals resample physical test cases, keeping their noise seeds and methods together. The same three noise seeds are reused across cases; these intervals describe case variation conditional on those common noise realizations, not uncertainty over a noise population or transfer to arbitrary sources or plants. The step-halving checks and all case outcomes remain in the fixed artifact package. Better fidelity, if observed, is not sufficient to establish calibrated source probabilities, all-family diagnosis or a benefit in the unchanged causal campaign.

D. HISTORY CONDITIONING IN AN INDEPENDENT REDUCED PLANT

A proposed extension conditions the evidence on the state left by a previous intervention. It relinearizes the signature of the new source at that state, adds the bilinear interaction between the two interventions, or integrates the uncertain

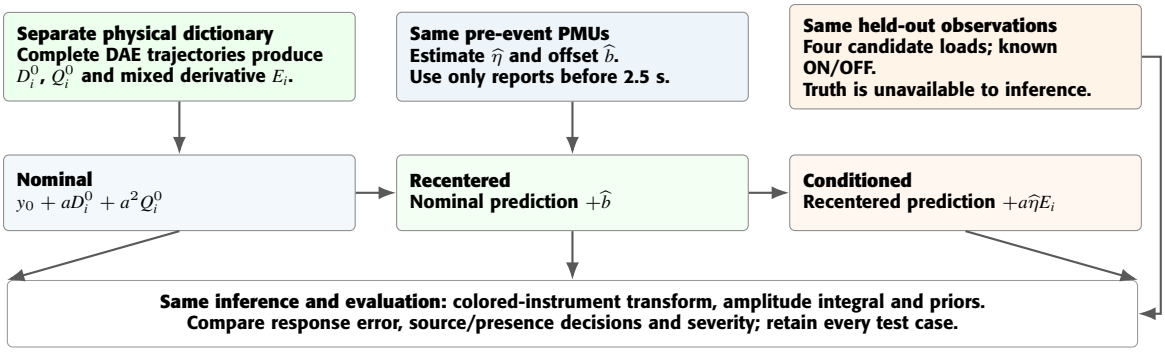


FIGURE 11. Executed signature ablation, separate from the causal stream algorithm in the main article. The three variants share the data and decision calculation; only reference and event-response terms differ.

TABLE 22. Controlled signature test, separate from causal detection. Response error is the median whitened incremental error at the generating source/severity over 80 physical cases. Joint correctness includes presence and source on 240 noisy event records; source-only ranking conditions on a load. Severity error is median absolute relative error at the generating source. Normal selections use 15 fixed windows, not normal-hours exposure.

Model	Response error (%)	Joint correct	Source-only correct	Normal event selections	Severity error (%)
Nominal	1.25	114/240	114/240	12/15	3.45
Recentered	1.25	240/240	240/240	0/15	1.2
Conditioned	0.0561	240/240	240/240	0/15	0.329

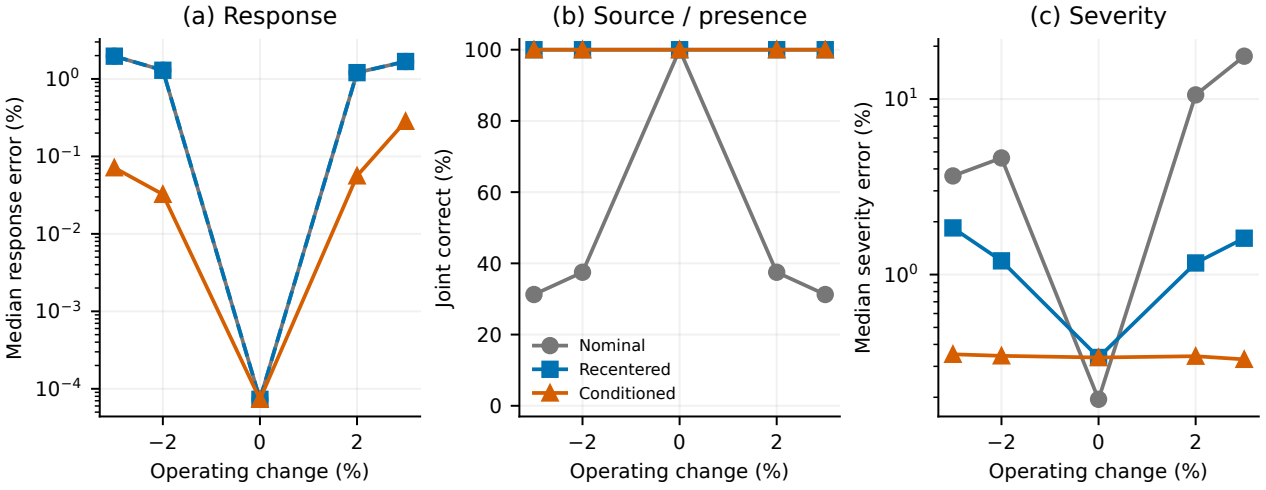


FIGURE 12. Separate effects of recentering and conditioning. (a) Incremental response error in the fixed physical test cases, evaluated at the generating source and severity to isolate approximation error. (b) Correct source and presence decisions on noisy observations, after marginalizing severity. (c) Relative severity error evaluated at the generating source. The panels measure different outcomes. Noise realizations repeat physical cases and are not counted as additional independent trajectories. Nominal and recentered incremental predictions coincide.

previous severity out of the likelihood. We tested it where it should matter most, with a plant different from the one used for the bank. It has classical machines, a 5% droop governor, constant-impedance loads and the network of the competition case. A load p of 8–16% of its base value is switched on at 2 s and off at 5 s; a weak second load of 2–6% appears at 6.5 s while the network is still recovering. The estimator knows p and the switching times, fits the previous severity by Gauss–Newton and ranks the new source among the eight loads of the catalog. Observation noise is the empirical covariance of the first differences of the competition record, multiplied by 1, 10 or 100. Some cells shorten the history available to the estimator or detune its inertia and damping by 20% and 40%. The protocol, the comparison criterion and the seeds were fixed before the test scenarios were generated, and each cell was run once with 200 scenarios.

Table 23 gives the outcome. In none of the 8 protocol cells does a predictor conditioned on the history make a significantly better source decision than the recentered nominal signature; the accuracies differ by at most 0.5 points. The mechanism is visible in the estimates. The previous severity is recovered to within 0.3 points of the base load in every pro-

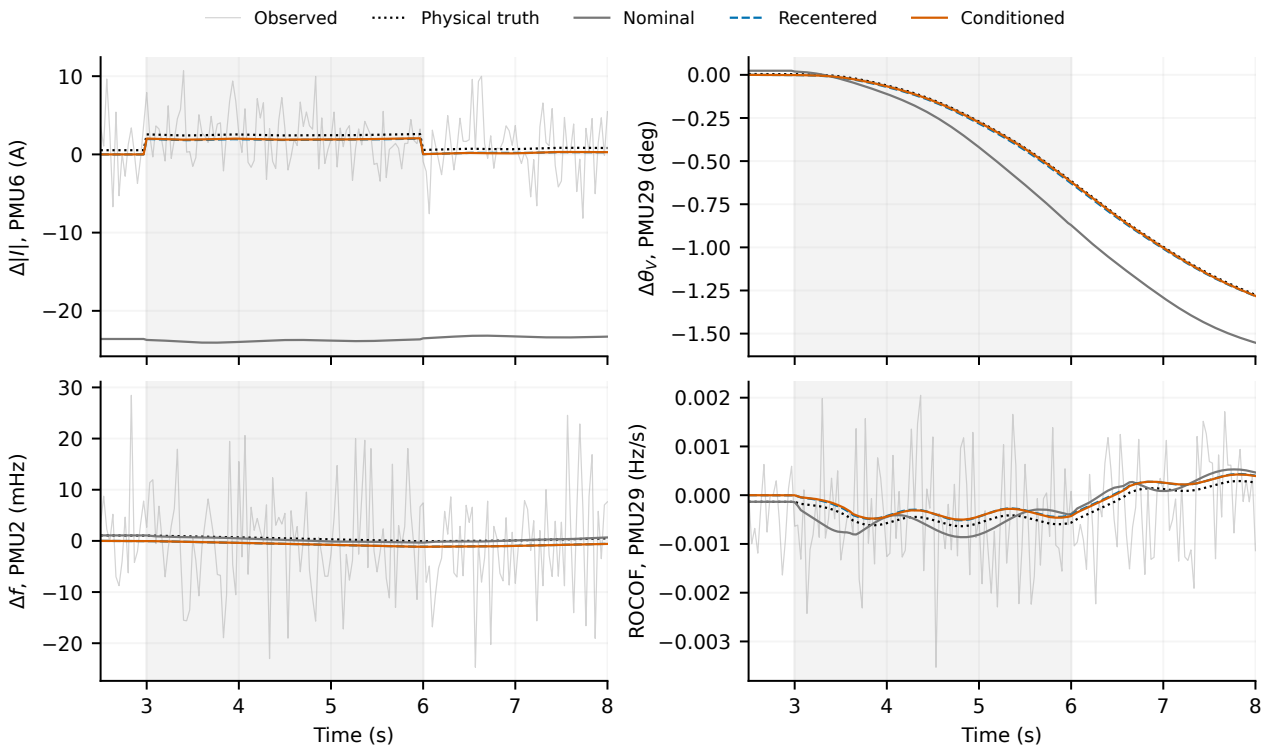


FIGURE 13. **Fixed illustrative test: BUS7, $\eta = +3\%$, load factor +1%, seed 390502. Gray traces are the delivered PMU observations. Colored curves use each method's selected source and estimated severity; a no-event selection keeps its baseline. The dotted black curve is physical truth, used only for illustration. The plotted channels and case were fixed without selecting the largest improvement. Shading marks physical ON/OFF for evaluation.**

tocol cell, so the history leaves no uncertainty to marginalize. We then looked for the regime where conditioning should finally pay off. Exploratory runs on separate seeds pushed the previous severity to 30–60%, started the new source 0.2 s after the previous one ended, detuned inertia and damping by 40% and 80%, and cut the history to 0.5–1 s with noise 100 to 1000 times the measured level. Even there the six predictors tied. We fixed two of those regimes (cells I and J) and ran them once on fresh seeds. The previous severity is then recovered only to 0.8–5.4 points, yet the decisions still do not change: the accuracies are equal in cell J and within one point in cell I, where integrating out the previous severity is even worse than the plug-in (by 1.0 point of accuracy and 0.9 nats of log-loss). The reason is structural. The state left by the previous intervention shifts the predicted record of every candidate source by almost the same amount, so it moves all hypotheses together and rarely reorders them; the ranking is set by the shape of the new source's own signature. The nominal signature costs 0.5 ms per decision, against 819 ms for the nonlinear plant and 6.4 s for the quadrature reference. A computable bound on the whitened prediction error certified the decisions of the nominal, bilinear and relinearized predictors in 78–99% of the scenarios. Among the certified decisions, 0 differed from the nonlinear plant. The bound needs the nonlinear prediction it bounds, so it validates the approximation and does not save computation.

On the competition record the same plant ranks the bus-7 load first among 16 load buses (rank 1) and explains 99.5% of the energy of the quasi-steady change after the label-free onset of the attributor. That record cannot separate the predictors, because its only load change arrives about ten minutes after the previous event has recovered. The extension would need a poorly observed previous intervention. A different plant could provide one, but none of the 8 protocol cells or the two extreme cells reached the point where conditioning changes a decision.

E. LEARNED BASELINES TRAINED ON THE BANK'S OWN SIMULATIONS

The ExtraTrees adapters of the main comparison keep the operating points of the earlier system. A fair test of the learned alternative retrains it where the bank was developed. We fitted three learners on the 24 training records of the simulated cohort (72 events) and fixed each alarm threshold on the 24 calibration records, requiring no alarm in normal records. Each was then scored once on the 24 held-out records with the same association rule. Inputs were causal. They were the changes of the 16 phasor channels over the last 0.5 and 1.5 s relative to a reference taken 1.5 to 4 s earlier, whitened with the startup noise of each record. One learner received those changes directly. Two received their projection on the 18 physical signatures of the catalog, which gives them the static physical knowledge of the bank. A multinomial logistic model on the projections serves as a linear control. The held-

TABLE 23. Source accuracy of five predictors in ten stress cells (200 scenarios each; mismatch is the relative error of the inertia, and the damping is detuned twice as much; cells I and J use a previous severity of 30–60%, a new source 0.2 s after the previous one ends, and fresh seeds). Nominal: equilibrium signature on the recentered baseline; plug-in: nonlinear plant at the estimated previous severity; marginalized: plug-in with the previous severity integrated out; quadrature: the same by seven-node quadrature. The last column is the paired accuracy difference marginalized minus plug-in with its 95% bootstrap interval, in percentage points.

Cell	Noise factor	History (s)	Mismatch (%)	Window (s)	Nominal (%)	Plug-in (%)	Marg. (%)	Quad. (%)	Marg.−plug-in (pp)
A	1	6.5	0	1.5	99.5	99.5	99.5	99.5	+0.0 [+0.0, +0.0]
B	100	6.5	0	1.5	98.0	97.5	98.0	98.0	+0.5 [+0.0, +1.5]
C	1	1	20	1.5	95.0	95.0	95.5	95.0	+0.5 [+0.0, +1.5]
D	100	1	20	1.5	93.5	93.5	93.5	93.5	+0.0 [+0.0, +0.0]
E	100	1	20	0.5	80.5	80.5	80.5	80.5	+0.0 [-1.5, +1.5]
F	10	6.5	0	1.5	99.5	99.5	99.5	99.5	+0.0 [+0.0, +0.0]
G	10	1	20	1.5	95.0	95.0	95.5	95.0	+0.5 [+0.0, +1.5]
H	100	0.5	0	1.5	98.0	98.0	98.0	98.0	+0.0 [+0.0, +0.0]
I	100	1	40	1.5	61.5	61.5	60.5	61.5	-1.0 [-2.5, +0.0]
J	1000	0.5	40	0.5	38.0	38.0	38.0	38.0	+0.0 [+0.0, +0.0]

TABLE 24. Learned baselines trained on the simulated cohort of the bank, scored once on its held-out 72 events (36 lines, 36 loads) with the same event-association rule. Thresholds were set on calibration records for zero alarms in normal records. Top-1 counts the events detected with the correct source. The 72 events come from 12 independent records of six events each, so the last column counts the records in which all six were located (family and source), with its exact 95% interval; the events of a record share one noise realization and operating point. Extras are alarms matching no event.

Method	Detected	Top-1	Lines	Loads	Extras	Records (95%)
Dynamic bank	72/72	72	36/36	36/36	7	12/12 (74–100)
Static physical	48/72	40	32/36	8/36	25	0/12 (0–26)
Trees, phasor changes	61/72	41	25/36	16/36	82	1/12 (0–38)
Trees, signatures	58/72	49	32/36	17/36	30	0/12 (0–26)
Logistic, signatures	55/72	32	27/36	5/36	71	0/12 (0–26)

out records had already been opened by the physical methods, so they are not a pristine test for those; they were not used to fit or to tune any learner.

Table 24 gives the outcome. The best learner locates 49 of 72 events at the source, against 72 for the dynamic bank, and the worst locates 32. Giving the learner the physical signatures helps most: it recovers 32 of 36 line outages but only 17 of 36 loads, where the responses are small and electromechanical and the static signatures describe them poorly. The learners also raise 30 to 82 alarms outside the scheduled events, against 7 for the bank. No learner raised an alarm in the normal records, by construction of the threshold. The result does not show that no learner can match the bank. It shows that with the data the bank itself was developed on, three standard learners, two of them given the physics as features, fall well short on loads.

F. TRANSFER TO A DIFFERENT PLANT

The dynamic bank has only been scored on one plant family, so a method could be good at that plant without being good at power systems. We re-simulated the 12 event schedules of the held-out cohort (same sources, instants and amplitudes) with an independent plant. It has classical machines, droop governors, constant-impedance loads and the competition network, and has no excitation system. The new records pass through the same PMU operator and the same instrumental noise, so only the physics differs. The voltages at the monitored buses match the original plant to within 0.1% at rest, while the response to a switching event differs by up to a factor of two. Four normal records of 120 s serve as controls. The exporter advanced one report before assigning the first timestamp, so these comparisons are exploratory and do not support timing or transfer claims. Every method was fixed before these records were read; learner thresholds were set on the original calibration records.

Table 25 gives the outcome. The static physical attributor is not affected by the change of plant: it locates 42 events against 40 on the original cohort. The learners lose up to 6 sources on the new plant and raise 75 to 273 alarms outside the scheduled events, several times more than before. Training on additional noise draws helps the phasor learner on the original cohort but not on the new plant, so this noise augmentation did not resolve the observed mismatch. No method locates all six events of any record, and no method raised an alarm in the normal records. The dynamic bank needs per-record banks built with the plant solver and has not been run on these records. This is the main gap of the evidence: the bank's 72 of 72 is measured on its own plant family, and a run on these records would bound how much of it depends on that match.

G. CLAIMS, EVIDENCE AND THREATS TO VALIDITY

Table 26 lists each headline claim with its evidence, the independent unit behind it and what it leaves open. The units matter. Replays of one physical parent share their trajectory, so the statistical weight of a result is set by the number of parents or records, not of events.

Plant family. The bank was developed and scored on simulations of one plant family. On an independent plant the static attributor keeps its accuracy and the learners lose up to 6 sources, but the bank itself was not run there. Its advantage over the other methods could shrink or grow on another plant, and this is the largest open point of the evidence. *Few independent units.* The held-out cohort has 12 event records

TABLE 25. Transfer to records simulated with a different plant (classical machines, droop governors, no excitation), built from the same 12 event schedules and passed through the same PMU operator and noise. Detected events, events located at the source (Top-1) and alarms matching no event, out of 72 events, for the original cohort and the new plant. The last column gives records with all six events located, out of 12; all methods raised zero alarms on the normal records. The dynamic bank needs per-record banks built with the original plant solver and was not run on these records. The two "more noise" learners were trained on four extra noise draws per training record.

Method	Original cohort			Second plant			Records
	Det.	Top-1	Extras	Det.	Top-1	Extras	(of 12)
Static physical	48	40	25	52	42	48	0
Trees, phasor changes	61	41	82	60	39	170	0
Trees, signatures	58	49	30	68	45	156	0
Trees, phasor changes, more noise	63	48	94	66	42	273	0
Trees, signatures, more noise	55	49	21	59	46	75	0

and the confirmatory simulation 18 parents. A method that is right in all 12 records is bounded below by 74% at 95% confidence, not by the event-level count. *Reuse of the held-out records.* They were opened by the bank, by two static attributors and, once each, by the learners of this section. No learner or threshold was tuned on them, but they are no longer a pristine test for the physical methods. *Competition record.* It is a validation set with logged iterations, and it contains one load change and one line outage among its twelve events. The bank does not start on it, so the competition record does not test the bank. *Event duration.* The simulated interventions last 2–5 s, while the labeled load changes of the record last minutes. The bank is evaluated where recovery dominates, and persistent changes are covered only by the static attributor. *Noise.* Independent per-PMU noise of 0.071% added RMS TVE degrades load attribution. Estimating cross-PMU covariance from the startup prefix is a candidate extension that was not tested. *Baseline strength.* The learners saw 72 training events, and more noise draws did not close the gap. A learner given far more physical variety could do better, so the comparison measures what the available data support.

APPENDIX Q RECOVERY GEOMETRY AND SOURCE SEPARATION

A. RESPONSE FIT, SEVERITY AND SOURCE IDENTITY

A retained physical history predicts the response still present after restoration and defines the state from which a new intervention begins. For the exact observation map Ψ and the same pre-intervention history h, the incremental response is

$$s_{e|h}(t) = \Psi_t(h \oplus e) - \Psi_t(h). \tag{38}$$

The two paths share their initial physical and instrumental states. In an additive linear time-invariant system, superposition makes this increment independent of the prior state. Nonlinear dynamics can change its shape, but do not guarantee greater source separation. Preserving recovery is a model-consistency requirement, not a claim that an earlier disturbance universally helps localization. The stream implementation approximates these paths with its declared switched predictors; it does not solve the full nonlinear DAE for each hypothesis.

A better physical prediction need not produce a better location. Competing sources may remain similar after their severities and initial conditions are fitted. For a response s_h at specified severity and context, the distance to an admitted competitor is

$$d_h^2 = \min_{j \neq h} \min_{\beta_j \in \mathcal{A}_j} \|s_h - s_j(\beta_j)\|^2_{\Sigma^{-1}}. \tag{39}$$

The set $\mathcal{A}_j$ fixes allowed signs, severities and nuisance coordinates. Small distance makes source discrimination noise-sensitive; it is not a universal error formula or a calibrated decision certificate [6], [7].

Three outcomes remain separate: prediction error at a specified physical source, ranking of competing sources, and uncertainty calibration. A response improvement establishes the first. It establishes the second only if source decisions improve under the same observations, horizon and decision rule. Neither establishes the third automatically. A nonlinear response comparison is needed before treating a surrogate's residual confusion as an unavoidable physical limit.

The separately compiled Historical Technical Notes retain the earlier state-reconstruction and simultaneous-load studies, with their separate observation contracts and prior-normalization qualification. Their known-onset scores are not combined with the causal results here. In particular, recovering hidden voltages does not certify the source that moved them.

To first order, a component of the context derivative parallel to a nominal source signature can be absorbed by fitting a different severity. Correct source ranking can therefore coexist with biased intensity. Components orthogonal to that signature cannot be removed by this rescaling. This explains why source accuracy can reach a test ceiling while response and severity errors still distinguish the models; it is a standard projection argument, not a new identifiability theorem.

B. DOES RECOVERY INCREASE SOURCE SEPARATION?

A separate physical control tests this possibility without changing the stream method. We solve 38 complete nonlinear trajectories of 8 s with 241 PMU reports each. A prior 8% load change at bus 7 or 24 is applied at 2 s and restored at 3 s. A weak 0.06% load change follows after 0.5 or 2 s. The candidate pairs are 18/27, 16/21 and 3/21, and the observation horizon is 1.5 s. Controls and prior-severity differences account for the remaining trajectories. Within each matched history, the secondary-event and no-secondary-event trajectories share the same prefix. Physical and instrumental states are retained through restoration.

This is an oracle component test: the prior trajectory, secondary onset, load family and candidate pair are supplied. Finite-amplitude differences of nonlinear trajectories provide incremental responses; these are secants, not newly derived analytical tangents. The 47-coordinate measurement operator is followed by its fixed innovations transform. For a geometric control, we profile nonnegative competing amplitudes.

TABLE 26. Claims, their evidence and their limits. The last column states what the evidence does not establish.

Claim	Evidence	Independent unit and size	Not established
The bank attributes load and line interventions before restoration	Independent simulation of the bank's plant family	18 parents, 108 interventions, 648 replays; 644/648 timely	Other plant families; the replays of a parent are not independent
Resetting the physical state at restoration costs attribution	Same simulation, paired arms	75 of 108 timely with reset against 108 of 108 with carry	Mechanism beyond the tested conditions
No false alarm in normal operation	Normal records of the same simulation	11.8 h, no alarm; a 95% bound of about 0.25 per hour	Field operation; unmodeled disturbances
Relations of sequence and network coupling diagnose missing and corrupted PMU episodes	Competition record	One record, all labeled episodes	A rate across records
A network-physics attributor locates every labeled event	Competition record, not blind 6 logged iterations on it	12 events in one record	A rate; a blind test
The bank attributes short pulses that the static attributor misses	Held-out simulated cohort	72 events in 12 records; 72 against 48 detected	A different plant (the bank was not run on one)
Learners trained on the bank's simulations fall short	Three learners, scored once	72 events in 12 records; at best 49 located	That no learner can match the bank
Conditioning on the previous intervention does not help	Reduced plant, 8+2 cells	200 scenarios per cell	The plant of the bank; a poorly observed history

Adding a rank-one covariance from uncertainty of 0.01 in the previous load fraction tests sensitivity to a nuisance direction; it is not a causal posterior over the full state.

The median ratio is 1.0006 with known history and 0.8532 with that nuisance direction; none of the 12 latter ratios exceeds one (Fig. 14). An illustrative binary likelihood comparison, using a signed Gaussian amplitude prior of standard deviation 0.0015 and 300 common Gaussian innovation draws per generating source, gives the same mean accuracy, 98.35%, for equilibrium and conditioned signatures. Two cells worsen and one improves. This likelihood experiment integrates signed amplitudes, whereas the distance profiles nonnegative amplitudes; the distance is not asserted to bound its error. The noise assumption is an innovations-domain control, not validation of the complete instrumental generator. These correlated, known-onset comparisons establish neither an improvement of the stream method nor general equivalence. They provide no support for claiming that recovery universally creates useful source information.

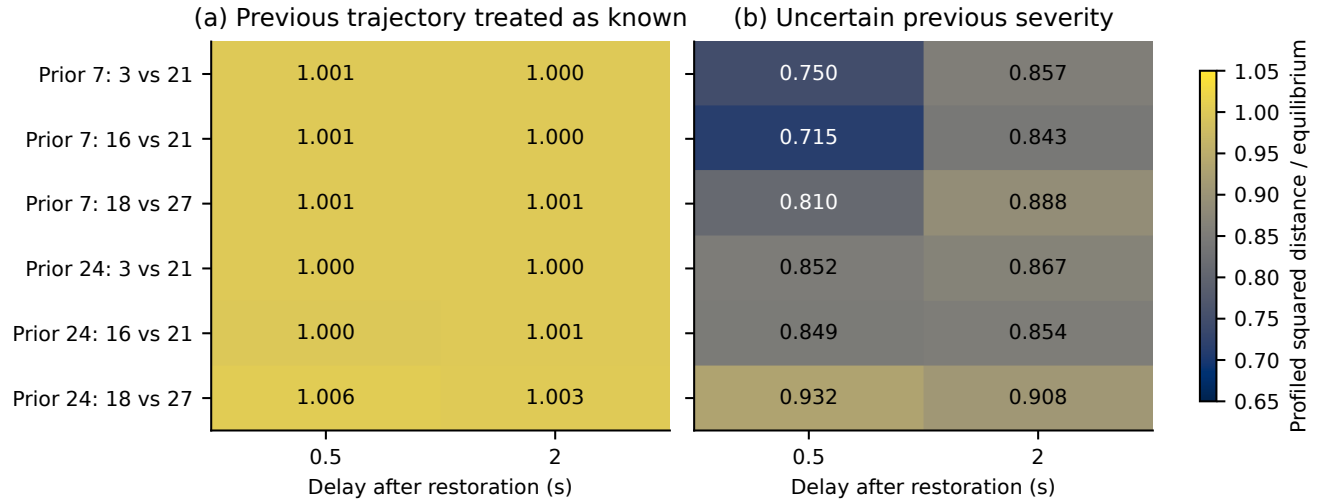


FIGURE 14. **Full nonlinear physical control, not a performance comparison of the stream method. Each cell is the profiled squared separation during recovery divided by its equilibrium value. The same source pairs are evaluated with the previous trajectory treated as known (left) and with one uncertain previous-severity direction (right). Values below one mean less separation. Adding nuisance covariance reduces precision at fixed signatures by construction; the left panel separately shows the small change in physical response geometry. The 12 cells share four recovery histories.**

APPENDIX R ARCHITECTURE AND A TESTABLE ADAPTIVE EXTENSION

This section specifies a successor and evaluates two of its mechanisms in nominal-operator diagnostics, not in the full history bank. Figure 15 separates the implemented physical estimator, refinement candidates, and an external cyber layer. Adaptive covariance estimation already appears in power-system dynamic state estimation, including an IEEE-39 study [8]. The question here is whether adaptation can follow instrumental changes without explaining away a load or line intervention during recovery.

A. CONDITIONAL FILTERING WITHOUT LOSING SEVERITY DEPENDENCE

For a fixed history, use the conventional predict–observe convention $\xi_k = F_h\xi_{k-1} + c_h + B_h a + w_k$ and $r_k = H_h\xi_k + d_h^{obs} + D_h a + v_k$. Assume independent zero-mean Gaussian process and observation noise, constant severity within the history, and covariance independent of severity. These timing conventions differ from the implemented combined one-report operators; the equations are not a drop-in replacement for that code. If $\xi_{k-1} \mid a, h, \mathcal{F}_{k-1} \sim \mathcal{N}(m^+ + d^+a, P^+)$, the prediction is

$$\begin{aligned} m^- &= F_h m^+ + c_h, & d^- &= F_h d^+ + B_h, \\ P^- &= F_h P^+ F_h^\top + Q_h. \end{aligned} \tag{40}$$

With $e = r_k - H_h m^- - d_h^{obs}$ and $s = H_h d^- + D_h$,

$$\begin{aligned} S &= H_h P^- H_h^\top + R_k, \quad K = P^- H_h^\top S^{-1}, \\ m^+ &= m^- + Ke, \quad d^+ = d^- - Ks, \\ P^+ &= (I - KH_h)P^-(I - KH_h)^\top + KR_kK^\top. \end{aligned} \tag{41}$$

The severity density is updated using $\mathcal{N}(e; sa, S)$ and the same normalized truncated-Gaussian integral as the main method. Its posterior variance V_a contributes $d^+V_a(d^+)^\top$ to the marginal state covariance. Substituting only the severity mean would lose this dependence. At fusion, within-history and between-history covariance must both be retained. Correlated process/observation errors require the corresponding cross terms; merely changing R in these equations is insufficient.

For an exogenous availability selector E_k, the admitted observation is $E_k r_k$, with operators $E_k H_h, E_k D_h$ and covariance $E_k R_k E_k^\top$. Missing values are marginalized, not replaced by zero or held as independent new reports. In particular, taking a submatrix of R_k^{-1} generally gives the wrong marginal precision. An informative, signal-dependent mask needs a selection model. The reported bank does not yet implement this general interface.

B. LEARN NOISE AFTER SCORING THE CURRENT REPORT

The first adaptive successor should fix Q, physical parameters and the observation center. In fixed scaled observation coordinates, use

$$\begin{aligned} R_k &= U\,\mathrm{diag}(\lambda_k)U^\top + \mathrm{diag}(\delta_k), \\ \lambda_k &\geq 0, \quad \delta_k \geq \delta_{\min} > 0, \end{aligned} \tag{42}$$

with fixed factor directions U, upper bounds and regularization selected on development data. Common reference and local sensor errors motivate these components; their factors are not freely identifiable in every PMU layout. Instrument transformations must act on the full covariance; temporally colored noise needs explicit states or a consistent whitening model, not an instantaneous R alone.

Let $\nu_r = r_r - \hat{r}_{r|r-1}$ be a pre-correction innovation and C_r^{pred} the nonmeasurement part of its predictive covariance. For a mixture, C_r^{pred} includes severity uncertainty and the

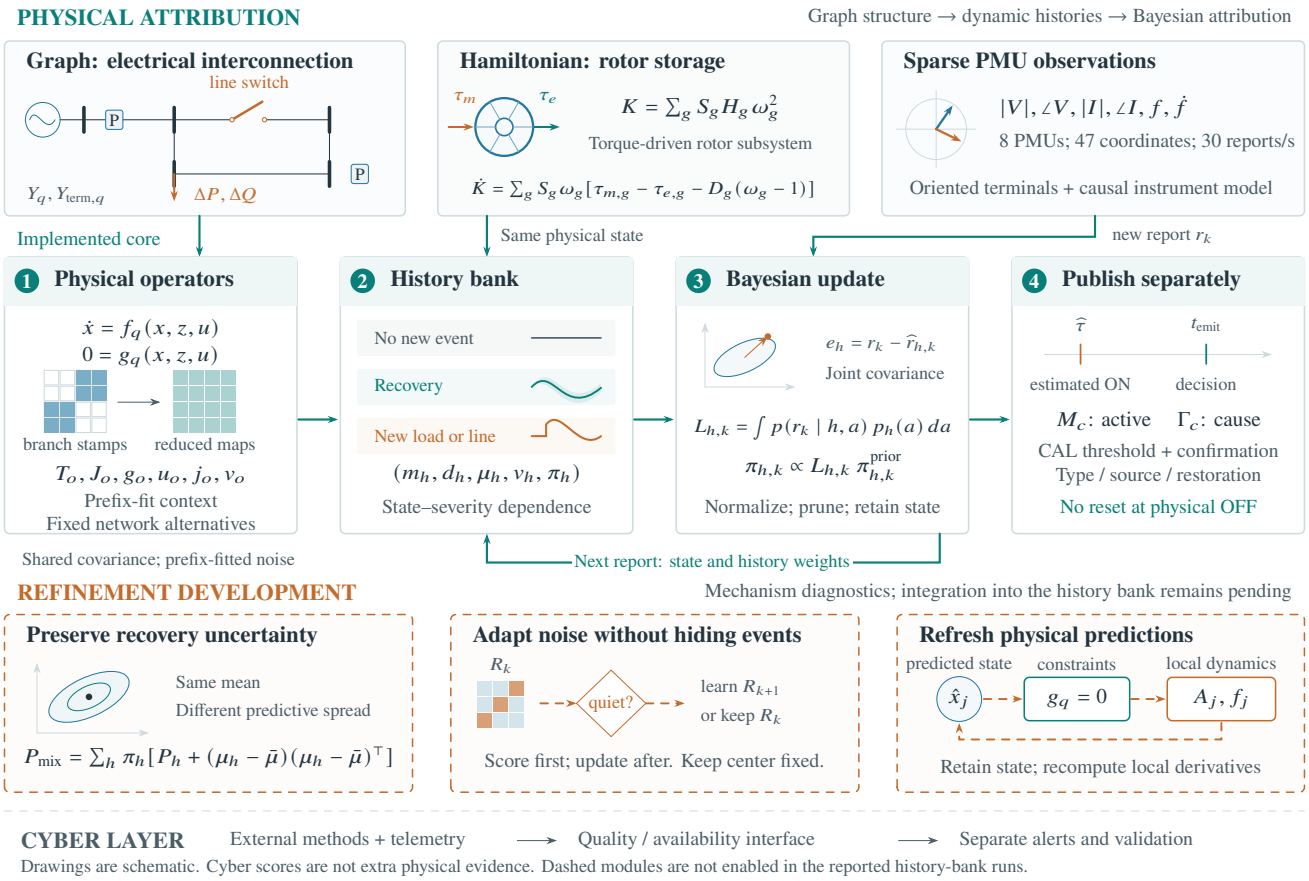


FIGURE 15. **System architecture and implementation boundary. The solid upper path is the reported physical bank. Dashed modules specify proposed integration of history-specific uncertainty, guarded noise adaptation and local dynamic refinement. The drawings are schematic; $p_h(a)$ denotes the severity density conditional on the history and past observations. Mechanism diagnostics below do not validate these modules within the bank. The separate known-event relinearization test appears in Section U. The rotor-energy panel describes a verified mechanical subsystem of the nonlinear plant (Section S); it contributes no additional likelihood factor. Graph connectivity enters through the electrical ports and mode-specific operators. The lower layer hosts separate cyber methods and outputs; neither cyber scores nor nominal source-separation diagnostics are extra physical likelihood factors. An availability interface requires a separately validated measurement model.**

spread of predictive means, not only $H_h P_h^- H_h^\top$. A candidate covariance-matching update is

$$\widehat{R}_k^{raw} = \frac{\sum_{r\in W_k} \omega_r (\nu_r \nu_r^\top - C_r^{pred})}{\sum_{r\in W_k} \omega_r},$$
$$R_{k+1} = \Pi_{\mathcal{R}}! \left[(1-\eta) R_k + \eta \widehat{R}_k^{raw}\right]. \tag{43}$$

If no eligible reports exist, retain R_k. Projection is a constrained fit to the positive-definite family, not unrestricted eigenvalue clipping. For example, bounded least squares on the fixed PSD factor matrices preserves that family. The identity $\mathbb{E}[\nu_r \nu_r^\top] = C_r^{pred} + R_r$ requires a correctly specified, conditionally centered predictor. A finite gated window need not satisfy it; bias or unmodeled recovery can inflate the estimated noise. Updating covariance must change both the quadratic score and its log determinant, as well as the gain. Score report k with R_k before allowing it to influence R_{k+1}.

Eligibility should depend on a sustained quiet-history state available before the report, independent quality metadata, and a declared recovery cooldown. This is an operational gate on approximate masses, not known event truth. Selection by the current residual is dangerous: for $Z \sim \mathcal{N}(0,1)$,

$$\mathbb{E}[Z^2 \mid |Z| < c] = 1 - \frac{2c\phi(c)}{2\Phi(c) - 1} < 1. \tag{44}$$

Thus retaining only small innovations underestimates variance even in a correct model. Past-only eligibility avoids this particular current-report truncation under the stated independence assumptions; it does not guarantee that a weak or sustained intervention will be recognized. The design must also test noise increases that make the gate stop admitting data.

Adapting the baseline can erase the event itself. If $b_{k+1} = (1-\eta)b_k + \eta a$, $b_0 = 0$, then before the next update $a - b_k = a(1-\eta)^k$. This elementary counterexample motivates keeping the center fixed in the first successor. Continuous impedance or operating parameter adaptation remains deferred until its effect can be separated from event severity; learning unrestricted Q, R and parameters together does not establish their identifiability.

C. HISTORICAL REVISION AND SEPARATE CYBER OUTPUTS

A bounded-window smoother needs a saved boundary distribution and a record of the covariance/parameter version used for each report. For the same boundary model, replace the old suffix by the recomputed one, $\log W_h^{new} = \log W_h^{old} - \ell_{h,W}^{old} + \ell_{h,W}^{new}$. Changing the boundary model instead requires rebuilding its contribution. Replaying a suffix must not rerun the adaptation learner or count its observations again. Revised historical estimates are published at the current report time; earlier alarms remain in the audit log. Unlike the current split inner/outer replay, a coherent successor must update both levels consistently or declare the remaining approximation. Plug-in $\widehat{R}_k, \widehat{\theta}_k$ do not constitute a joint Bayesian posterior over those parameters.

The external cyber layer consumes its own telemetry and metadata and emits separately evaluated alerts. Missing or corrupted PMU reports do not establish a cyber attack. A quality flag may enter a declared measurement-selection rule; an attack score is not another independent likelihood of the same phasors. Simultaneous physical and integrity changes require paired tests of that interface.

D. VERIFICATION AND EVIDENCE STILL REQUIRED

The bundled numerical verification checks the affine update and severity integral against numerical quadrature, the Joseph covariance against Gaussian conditioning, structured covariance projection, missing-coordinate marginalization, prefix invariance and suffix replacement on a small adaptive Gaussian model. It also verifies the two counterexamples above. These are numerical checks of the proposed building blocks, not load-attribution or false-alarm results.

E. NOMINAL-OPERATOR MECHANISM DIAGNOSTICS

Two independent diagnostics use the fixed nominal IEEE-39 load operators with nine load sources and 47 observation coordinates. They do not use observations from the mechanism/transfer campaign. The code, noise seeds, threshold grid and endpoints were fixed before generating the cohorts. This corrected study replaces a withdrawn diagnostic that used a mid-report ON operator at restoration. The corrected OFF index is checked independently against the operator-builder ordering; all noise seeds are new and no other method settings were tuned. The deterministic null-innovation recursion reproduces explicit physical state and null-filter propagation through ON and OFF to maximum absolute error 2.8×10^{-10}. Noise is generated in innovation coordinates; neither a nonlinear second plant nor independent per-PMU TVE is simulated.

a: Inherited uncertainty at a fixed recovery mean.

The first diagnostic holds the onset, current-event amplitude and recovery mean known. Over 15 reports, let μ_c be the nominal signature of source c and let u be the preceding load's recovery signature scaled by its severity standard deviation, 0.002. In whitened coordinates the fixture is

$$y = \mu_c + u\zeta + \epsilon, \qquad \zeta \sim \mathcal{N}(0,1), \quad \epsilon \sim \mathcal{N}(0, I), \tag{45}$$

with independent ζ, ϵ and dimension $15\times47 = 705$. The preceding load is restored 0.5 s before the observation window. The two source selectors use the same means and respectively covariance I or $I + uu^\top$. The latter evaluates

$$d_c = \|y-\mu_c\|^2 - \frac{[u^\top(y-\mu_c)]^2}{1+u^\top u},$$
$$\log\det(I+uu^\top) = \log(1+u^\top u). \tag{46}$$

The covariance is common across candidate sources for each draw, so its log determinant cancels in the source ranking, but remains in the density. Across 360 independent noise/severity draws, correct source identification increased from 279/360 to 343/360: $+17.78$ percentage points, paired 95% bootstrap interval $[13.89, 21.94]$ from 10,000 resamples of the draws. Coverage of the nominal 95% true-source innovation ellipsoid increased from 62.22% to 96.11%. This demonstrates the cost of discarding a known covariance component under the declared Gaussian model. It neither compares explicit histories with an equivalent fused distribution nor establishes detector performance with unknown onset and amplitude.

b: Causal noise learning with an unchanged observation center.

The second diagnostic adds covariance $\alpha_k \operatorname{diag}(S_0)$ to nominal innovation covariance S_0. Normal records contain stationary, step-changing or ramping noise. Each event record contains two nonoverlapping loads, active at 30–70 s and 80–90 s, with magnitudes 0.0005 or 0.002 and both signs. A per-report source/age mixture compares nine sources and eight fixed ages with a Gaussian severity prior of standard deviation 0.01. It is a local likelihood scan, not a temporal history posterior, and uses an untruncated severity prior. No event-type or restoration estimator is run.

The arms hold $\alpha = 0$, learn continuously after startup, or learn only after 15 quiet reports and a 60-report cooldown. Quiet means nominal posterior mass at least 0.99 under normal prior mass 0.99. The noise moment estimate uses a learning rate $1/900$ and projects α onto $[0, 8]$. Each report is scored before learning; the current report can affect the guard only for later reports. All arms share the same likelihood scan, six-report publication confirmation and five-second refractory interval. The lowest feasible threshold on the fixed grid is selected using only 36 normal calibration records, each 600 s with 12 s startup. Thresholds are fixed before generating 36 new normal and 36 event TEST records. Publication does not modify inference in this diagnostic.

Table 27 gives the held-out outcomes. Continuous versus fixed adaptation changes timely source coverage by 0.00 percentage points, with a paired 95% interval $[-8.33, 6.94]$ from 10,000 resamples of the 36 event records. The guarded and fixed arms identify the same number in every record. Thus, no attribution gain is established. Fixed and continuous learning also exceed the one-alarm-per-hour target on TEST; meeting

TABLE 27. Nominal innovation diagnostic: independent calibration at one normal alarm per hour. All TEST arms use 5.88 normal hours and 72 load interventions in 36 paired records.

Noise rule	h	Correct/72	Extras	FAR (upper)
Fixed	3	20	44	1.02 (2.01)
Continuous	0	20	50	1.36 (2.45)
Guarded	3	20	44	0.85 (1.79)

FAR and its one-sided 95% Poisson upper bound are in h^{-1}. Extras include repeated or incorrect publications, over 1.08 monitored event hours. Correct means source identified during its true ON interval; this diagnostic estimates neither event family nor OFF.

that target in calibration did not ensure it on new records. These outcomes are not a comparison at equal achieved false-alarm rates. Normal-record noise-coefficient RMSE is 0.875, 0.232 and 0.872 for fixed, continuous and guarded learning. The guard admits only 8.34% of monitored normal reports: causal timing alone does not prevent a gate from obstructing adaptation to noise changes. These settings were not retuned after TEST. The exact Poisson limits in the table are conditional on that count model; the heterogeneous noise schedule and refractory rule do not establish a homogeneous Poisson process.

The deterministic prefix check passes for all inference arrays. An independent recount verifies the saved prediction streams, calibration selection, event matches, normal exposure and saved-file integrity. These diagnostics support preserving inherited uncertainty and revisiting the adaptation guard on new development data. They do not support adding an adaptive-performance claim to the article abstract.

A fresh full-bank campaign should first compare fixed shared covariance with history-specific covariance, then add structured noise adaptation with and without the history gate. Test smoothing and memory compression separately. Each arm needs independent calibration to the same normal-alarm target, with infeasible arms declared. Include independent PMU noise, normal noise changes, long weak interventions, recovery, parameter shift and missing reports. Freeze settings and parents before TEST; compare timely attribution, extra alarms, normal exposure, predictive score, availability and latency by parent. Neither this design nor its numerical checks modify the ongoing campaigns or establish superiority of the adaptive extension.

APPENDIX S GRAPH INTERCONNECTION, ROTOR ENERGY, AND BAYESIAN EVIDENCE

The graph enters through electrical operators. The plant has 114 differential states, including ten Sauer–Pai machines and controls; its augmented predictor has 376 coordinates. Rotor angle δ_g, per-unit absolute speed ω_g and the deviation vector Δx are distinct. PMU frequency comes from voltage phase, not rotor speed.

A. AN EXACT MECHANICAL SUBSYSTEM BALANCE

The implemented machine equations, with frame speed one, are

$$\begin{aligned}\dot{\delta}_g &= \omega_b(\omega_g - 1),\\ 2H_g\dot{\omega}_g &= \tau_{m,g} - \tau_{e,g} - D_g(\omega_g - 1).\end{aligned} \tag{47}$$

Here $\tau_{m,g}, \tau_{e,g}$ are per-unit torques on machine rating S_g; ω_b is in rad/s and H_g in seconds. Multiplying the second equation by $S_g\omega_g$ gives the exact kinetic balance

$$\begin{aligned}K &= \sum_g S_g H_g \omega_g^2,\\ \dot{K} &= \sum_g S_g\omega_g[\tau_{m,g} - \tau_{e,g} - D_g(\omega_g - 1)].\end{aligned} \tag{48}$$

With ratings expressed numerically in MVA on the model's power bases, K is in MJ and its derivative in MW. The algebraic stator equations also imply, on each machine base,

$$\omega_g\tau_{e,g} = P_g + R_{s,g}(I_{d,g}^2 + I_{q,g}^2). \tag{49}$$

Replacing torque by terminal power without this speed factor would change the implemented dynamics. Governor torque and electromagnetic torque remain ports of this mechanical subsystem; their evolution is not discarded.

A canonical description of the rotor deviations follows by defining

$$\begin{aligned}p_g &= \frac{2H_gS_g}{\omega_b}(\omega_g - 1),\\ \mathcal{H}_{rot}(p) &= \sum_g \frac{\omega_b^2 p_g^2}{4H_gS_g}.\end{aligned} \tag{50}$$

Then $\dot{\delta}_g = \partial\mathcal{H}_{rot}/\partial p_g$ and

$$\dot{p}_g = \frac{S_g}{\omega_b}(\tau_{m,g} - \tau_{e,g}) - \frac{S_gD_g}{\omega_b^2}\frac{\partial\mathcal{H}_{rot}}{\partial p_g}. \tag{51}$$

This is a Hamiltonian rotor subsystem with torque ports and nonnegative damping when $D_g \geq 0$. In this fixed plant direct shaft damping is zero; controller dynamics are still present. Neither K nor $\mathcal{H}_{rot}$ is a total energy or a Lyapunov function for the complete lossy DAE. A global port-Hamiltonian realization would require compatible electrical storage and controller interconnections [2].

B. THE GRAPH CONNECTS ELECTRICAL POWER PORTS

Let C_f, C_t select the source and destination buses of each branch, including the declared transformer orientation. With outward terminal currents,

$$\begin{aligned}i_f &= Y_{f,q}V, \quad i_t = Y_{t,q}V,\\ i_{\mathrm{net}} &= C_f^\top i_f + C_t^\top i_t.\end{aligned} \tag{52}$$

The branch and nodal active powers obey the algebraic identity

$$\Re(V^\top\overline{i_{\mathrm{net}}}) = \Re[(C_fV)^\top\overline{i_f} + (C_tV)^\top\overline{i_t}]. \tag{53}$$

Separately modeled bus shunts must be added if present. The actual branch stamps retain resistance, charging and

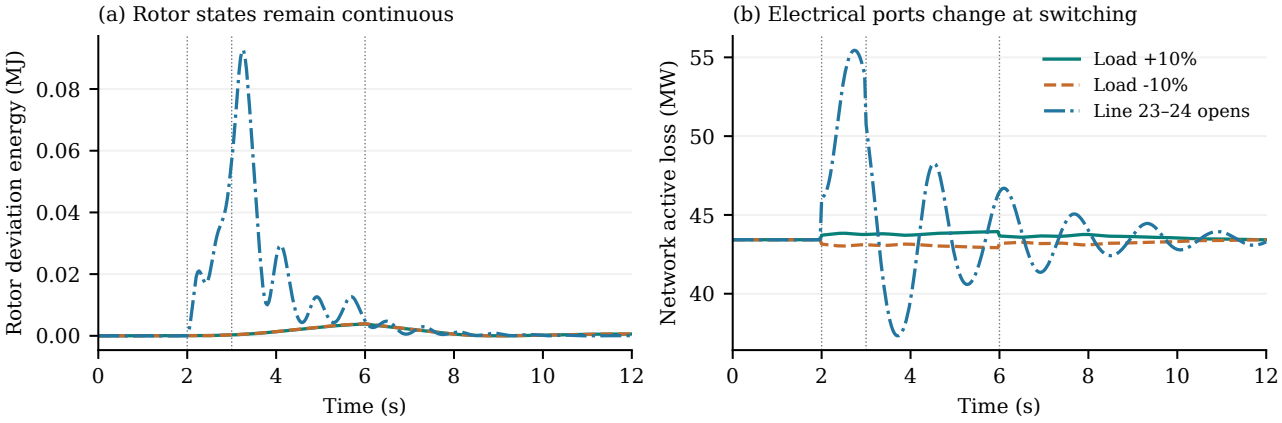


FIGURE 16. **New nonlinear IEEE-39 consistency trajectories. Changes begin at 2 s; the line is restored at 3 s and loads at 6 s. The normal trajectory has zero rotor deviation energy and is omitted for clarity. Rotor energy uses simulated internal states, not eight-PMU measurements. The traces verify the mechanical/electrical connection; they do not establish an energy-based event detector.**

transformer ratios; a simple lossless Laplacian is insufficient. Opening a branch changes these ports and the algebraic solution while differential states, hence rotor energy, remain continuous. The ports affect electrical torque, which then changes rotor motion. This connects graph structure to recovery without introducing an independent energy observation.

C. EXECUTED PHYSICAL AND INFERENCE CONTROLS

Four new 12-s trajectories use the unmodified nonlinear plant and initial condition: normal, a $+10\%$ and a -10% change of the bus-7 impedance load's P/Q setpoints from 2 to 6 s, and opening line 23–24 from 2 to 3 s. Actual load power remains voltage-dependent. All four solves succeed, with zero detected differential-state jump. At 120 samples/s the largest rotor-equation residual is 1.01×10^{-15} and the largest generation–load–network-loss mismatch is 4.13×10^{-11} system pu. Numerically integrating mechanical work separately on smooth segments gives maximum kinetic-energy discrepancy 1.06×10^{-2} MJ; intervals ending at a switching instant are excluded from trapezoidal integration. These are physical consistency checks, not independent statistical replicates or event-detection results.

To test whether the energy coordinates themselves add inferential value, let $\eta = L\xi$ be their invertible linear scaling of deviation coordinates. Every operator must be transformed consistently:

$$\begin{aligned} T_q' &= LT_qL^{-1}, & J_q' &= J_qL^{-1}, \\ g_q' &= Lg_q, & u_q' &= Lu_q, \\ P_h' &= LP_hL^{\top}, & K_h' &= LK_h. \end{aligned} \tag{54}$$

Transform means, severity directions and process covariance likewise; observation offsets and noise stay unchanged. Predicted reports and innovation distributions then coincide. Transforming the state prior with its Jacobian also preserves the integrated likelihood. Thus an exact Bayesian comparison with the same histories and publication rules gains no information. Coordinate-dependent approximations need a separate check.

Using all 18 nominal source operators through ON, stay and OFF, the maximum relative prediction discrepancy is 1.53×10^{-14}. In 180 paired Gaussian known-signature draws, native and rotor-coordinate likelihoods select the same source every time. The largest absolute log-likelihood discrepancy, 1.91×10^{-6}, exceeds the preset 10^{-6} numerical gate; this gate remains failed. The result is reported as a finite-precision limitation, not as an all-checks-pass certification. This control does not rerun the bank's proposal search, pruning or unknown-severity integration. It supplies no evidence of improved attribution.

Energy reconstructed from the same PMUs is not independent evidence. An energetic prior or discrepancy constraint would change the model and require a new comparison. The validated result is a mechanical interpretation and its graph coupling; no additional event score or accuracy gain is established.

APPENDIX T FROM THE NONLINEAR PLANT TO EXECUTABLE PREDICTORS

The reduction in the method is checked against the implemented plant, rather than a separate classical-machine example. This development diagnostic uses the fixed initial state and parameter vector, before prefix fitting, with 192 native variables: 114 differential and 78 algebraic. It does not rerun the conditional bank or estimate an event from PMUs.

A. MASS NORMALIZATION AND DERIVATIVE CHECKS

Write the native residual as

$$M_d\dot{x} = F_q(x,z,a), \qquad 0 = G_q(x,z,a), \tag{55}$$

with constant nonsingular differential mass block M_d. At the same differential reference $\bar{x}$ in each topology, solve

$G_q(\bar{x}, \bar{z}_q, 0) = 0$. Automatic differentiation of the assembled network gives the blocks used in

$$\begin{aligned} Z_x &= -G_z^{-1} G_x, & Z_a &= -G_z^{-1} G_a, \\ A_q &= M_d^{-1}(F_x + F_z Z_x), & B_q &= M_d^{-1}(F_a + F_z Z_a), \\ c_q &= M_d^{-1} F_q(\bar{x}, \bar{z}_q, 0), & C_q &= H_x + H_z Z_x, \\ D_q &= H_a + H_z Z_a. \end{aligned} \tag{56}$$

Thus the main text's f_q is the mass-normalized differential field. The reference is fixed; a moving reference would additionally contribute $-\dot{\bar{x}}$ and require consistent state and output recentering. Here H_q first selects rectangular bus voltages. Topology-specific terminal stamps and polar derivatives form the PMU observations. The load input a scales the bus-7 P/Q setpoints together. Because its load law is constant impedance, this is not a prescribed change in instantaneous consumed power.

An independent numerical reconstruction of all four reduced operators agrees to 4.3×10^{-14} in maximum absolute entry error. Centered directional checks against the native residual and consistent nonlinear algebraic solves have maximum relative discrepancy 3.91×10^{-8}, below the fixed 10^{-5} gate. These checks cover the intact network and line 23–24 open. They establish local derivative consistency, not accuracy for a finite excursion. The algebraic Jacobian condition numbers are approximately 2109 and 2112 in the native scaling; neither value is a coordinate-invariant margin.

B. OPEN-LOOP TRAJECTORIES AND OBSERVATION COORDINATES

Six new 12-s nonlinear trajectories use the same initial condition: normal, bus-7 setpoint changes of $\pm 1\%$ and $\pm 10\%$ from 2 to 6 s, and line 23–24 open from 2 to 3 s. All solves succeed. Samples include both 0 and 12 s; reports at switches use the post-switch algebraic state and preserve the differential state. Terminal predictions at the eight declared PMUs match the native current observables to 4.52×10^{-13} pu, retaining orientation, transformer stamps and the provisional bus-5 terminal assignment.

The reduced models start at zero deviation and propagate by an augmented matrix exponential at 30 reports/s. They know the event times, location and magnitude. No measured state correction, noise, covariance adjustment or parameter fit is applied. At restoration, the state carries forward and the nominal operators resume. The comparison therefore tests the construction and its finite-excursion error; it does not measure source attribution or the accuracy of the augmented conditional bank.

For $\hat{V} = \bar{V} + \Delta V$, distinguish applying an exact polar conversion to an approximate complex voltage from linearizing the polar observation itself:

$$\begin{aligned} \widehat{|V|} &= |\bar{V}| + \Re\left(\Delta V \frac{\bar{V}^*}{|\bar{V}|}\right), \\ \hat{\theta} &= \arg \bar{V} + \Im(\Delta V / \bar{V}). \end{aligned} \tag{57}$$

TABLE 28. Open-loop errors at eight PMUs. Relative complex-voltage RMSE is normalized by the nonlinear event-response RMS; magnitude and phase errors use the first-order polar observation during recovery.

Event	Complex error (%) ON	Complex error (%) Recovery	Polar RMSE after OFF $\lvert V\rvert$ (pu)	Polar RMSE after OFF θ (deg)
Load $+1\%$	0.72	1.48	1.82×10^{-8}	0.0012
Load -1%	0.72	1.48	1.82×10^{-8}	0.0012
Load $+10\%$	7.14	14.82	1.82×10^{-6}	0.1189
Load -10%	7.17	14.89	1.82×10^{-6}	0.1193
Line 23–24	17.97	30.70	7.78×10^{-4}	3.1910

The phase expression is in radians. Current channels use the same derivatives at their nonzero reference currents, with the declared terminal polarity and instrument gains. Both maps have the same local derivative, but differ over a finite excursion. For example, $\Delta V = \mathrm{j}\bar{V}\epsilon$ describes a phase tangent. Taking its exact magnitude yields $|\bar{V}|\sqrt{1+\epsilon^2}$, an artificial second-order magnitude rise. The linearized magnitude correctly has zero first-order response. Figure 17 shows why a small-signal complex-voltage approximation should not be interpreted as an exact finite-rotation phasor.

Table 28 separates complex-voltage error from the polar observation errors. Recovery means 6–12 s for loads and 3–12 s for the line, including the restored report. Reducing the load step by ten reduces its absolute complex-voltage error by about one hundred. This is consistent with local first-order accuracy in these trajectories, not a global approximation bound. The line's recovery phase error remains about 3.19 degrees with first-order polar observations; correct derivatives do not remove finite-event model mismatch.

The reproducible outputs also retain separate magnitude, phase, frequency and ROCOF errors for all 47 admitted coordinates. One comparison applies the same robust causal phase estimator to both phasor trajectories; another uses first-order polar channels and the predictor's ordinary 21-report polynomial fit. Their frequency and ROCOF errors include different observation approximations and are not interchangeable. These six deterministic cases have no independent statistical replicates. They add a construction check and an explicit approximation limit, not a Hamiltonian, graph-learning or adaptive-detection accuracy claim.

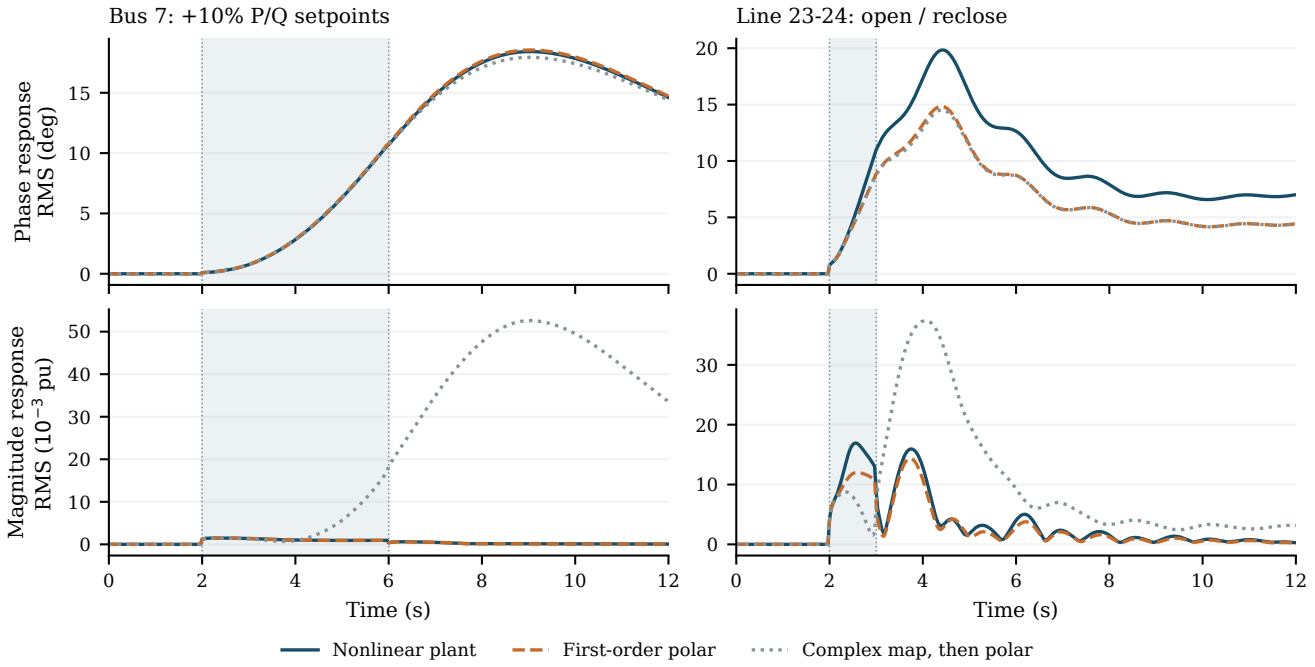


FIGURE 17. **Open-loop predictions from the actual plant derivatives, with known intervention times and no filtering. Curves are RMS deviations from the matching normal trajectory over eight PMUs, not RMSE; shaded intervals mark the active intervention. Solid: nonlinear plant. Dashed: reduced dynamics with first-order polar observations. Dotted: reduced complex voltage followed by exact polar conversion. The latter creates a large spurious magnitude response during load recovery. The line case retains a phase error with either map. Table 28 reports errors for both load signs and magnitudes.**

APPENDIX U REDUCING FINITE-EVENT PREDICTION ERROR

The preceding line-recovery error motivates a separate predictor comparison. The hypothesis is that recomputing local dynamics along the predicted trajectory reduces the error left by a fixed expansion point. This changes the physical approximation, not the plant, terminal assignment or noise law.

At a refresh time t_j, take the predictor's current state as $x_j = \hat{x}(t_j)$ and solve $G_q(x_j, z_j, a) = 0$. Compute the mass-normalized field f_j and reduced Jacobian A_j there. Until the next refresh, propagate the local affine model by

$$\hat{x}(t_j + s) = x_j + \left[\exp\left(s \begin{bmatrix} A_j & f_j \\ 0 & 0 \end{bmatrix}\right)\right]_{1:n,n+1}. \tag{58}$$

This expression starts from zero deviation at each new anchor, retains the absolute predicted state, and requires no inverse of A_j. At every report, solve the algebraic constraints at $\hat{x}$ and form the topology-specific voltage and terminal-current observations. A known switch forces an immediate refresh; differential states are not reset. The new reference comes entirely from previous model predictions, never subsequent true internal states or measured corrections. This is local linearization with nonlinear algebraic observations, not an iterated measurement update.

A. SEPARATE DYNAMIC AND OBSERVATION REFINEMENT

The protocol fixes a refresh every three reports (0.1 s) before execution. It compares the fixed dynamics and first-order polar observation of the preceding section, the same fixed dynamics with nonlinear algebraic observations, and the refreshed dynamics with those same observations. A one-report refresh is retained only as a convergence diagnostic. No interval is selected by minimizing the observed errors.

TABLE 29. **Recovery voltage-phase RMSE (degrees). All methods receive the same known event and initial state. The 30-Hz refresh is a convergence check, not a replacement selected after evaluation.**

Event	Fixed polar	Nonlinear output only	Relinearize 10 Hz	Relinearize 30 Hz
23–24, 1 s	3.191	3.191	0.139	0.0155
23–24, 0.5 s	2.542	2.541	0.246	0.0279
23–24, 1.5 s	3.390	3.390	0.181	0.0203
26–28, 1 s	3.211	3.212	0.150	0.0169
Bus 7, +10%	0.119	0.119	6.6×10^{-5}	7.5×10^{-6}
Bus 7, −10%	0.119	0.119	6.6×10^{-5}	7.5×10^{-6}

Seven 12-s trajectories comprise a normal control, the previous bus-7 ±10% load changes, the previous 1-s opening of line 23–24, two new opening durations of 0.5 and 1.5 s on that line, and a 1-s opening of line 26–28. Each event starts at 2 s; loads restore at 6 s. All methods know the event, its times and the initial condition. All seven nonlinear solves and all predictor runs succeed, with continuous differential states and algebraically consistent outputs. These are deterministic development checks with three prospective line cases, not independent random parents or a transfer experiment.

The reference line's recovery phase RMSE falls from

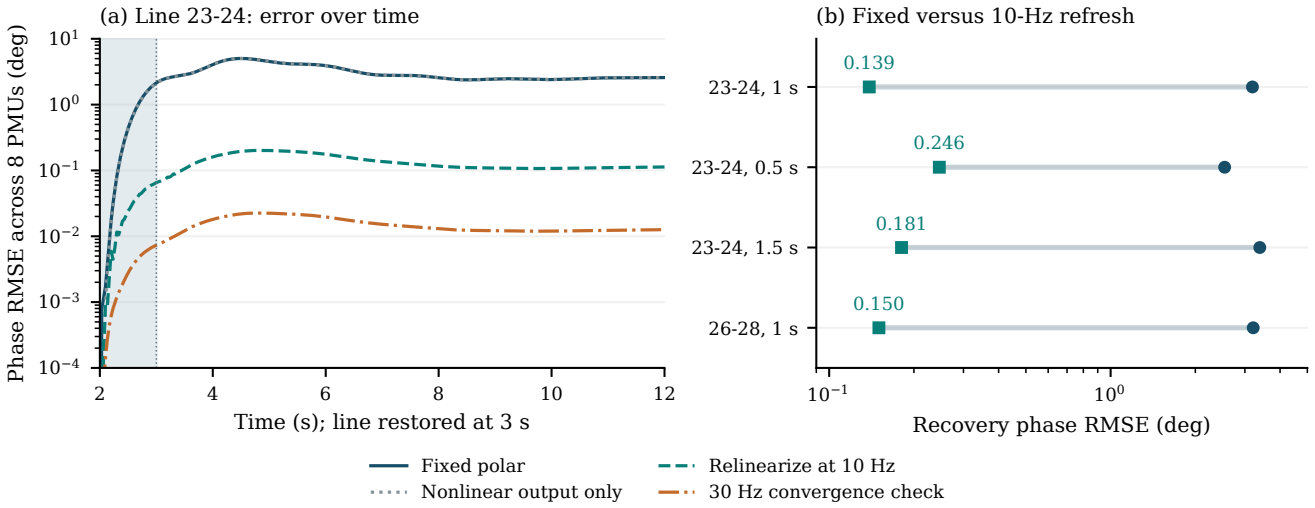


FIGURE 18. Prediction refinement with known events and no measurement updates. (a) Phase RMSE across eight PMUs at each report; the shaded interval marks the active opening. The observation-only control almost overlaps the fixed predictor. (b) Recovery RMSE over each case's restored interval: circles use fixed dynamics, squares refresh every three reports. All four line cases are shown. The finer refresh in (a) checks convergence; it is not a method selected after inspecting the comparison.

3.191° to 0.139°, a 95.7% reduction over eight PMUs from restoration through 12 s. Changing only the output map leaves the error essentially unchanged (Table 29); refreshing the dynamics supplies the improvement. Every line case exceeds the preset 50% reduction criterion. Both load controls satisfy the preset no-regression criterion, and the normal control remains within 10^{-8} pu of its nonlinear voltage trajectory. Refreshing each report further reduces the reference line error to 0.0155°, consistent with remaining discretization error. It is not the selected primary result.

B. WHAT THIS ENABLES, AND WHAT REMAINS UNTESTED

The result supports refining physical trajectories when a fixed local model is inaccurate. It does not establish improved attribution: event times and sources were given, measurements were not assimilated, and state covariance was not propagated. The saved diagnostics report all 47 channel errors separately, preserving units. Frequency and ROCOF still compare the predictor's ordinary causal polynomial fit with the plant's robust report estimator. Timings cover only this physical predictor and exclude bank search, uncertainty and report formation; they are not a real-time benchmark of the detector.

A bank successor would need a derivative of its chosen state-dependent prediction map, a compatible uncertainty recursion, and pre-correction predictive likelihoods. Nonlinear severity dependence also prevents assuming that the current analytic Gaussian severity integral remains exact. A conditional quadrature or another explicitly checked approximation would be required. An iterated measurement update is a distinct extension; recursive DAE state estimators already have relevant antecedents [9]. Iterations must retain the same prior and assimilate each report once, and a fitted residual alone is not marginal evidence. The reported history bank and its existing campaigns remain unchanged by this predictor-level result.

C. CONDITIONS FOR SELECTIVE REFINEMENT

A smaller trajectory error need not preserve attribution. Consider a reference Gaussian prediction (μ, Σ) and an approximation $(\widetilde{\mu}, \widetilde{\Sigma})$ of dimension d, both with positive definite covariance. Choose $W\widetilde{\Sigma}W^\top = I$, and suppose valid bounds satisfy

$$\begin{aligned} \|W(\mu - \widetilde{\mu})\| &\le \epsilon, \\ \|W\Sigma W^\top - I\|_2 &\le \eta < 1. \end{aligned} \tag{59}$$

Writing $r = \|W(y - \widetilde{\mu})\|$, the log-density error obeys

$$|\ell - \widetilde{\ell}| \le B = \frac{1}{2}\left[-d\log(1-\eta) + \frac{\eta r^2 + 2r\epsilon + \epsilon^2}{1-\eta}\right]. \tag{60}$$

Indeed, the whitened covariance has eigenvalues in $[1-\eta, 1+\eta]$. Bound its log determinant and inverse, then expand the quadratic residual. For unchanged covariance this reduces to $B = r\epsilon + \epsilon^2/2$; a mean-only bound is insufficient when covariance changes.

For a fixed candidate set and the same priors, bounds B_h preserve the approximate score winner i if

$$\widetilde{S}_i - B_i > \max_{j \ne i}(\widetilde{S}_j + B_j). \tag{61}$$

This condition does not preserve a publication threshold on normalized event mass. For an event group, sum the exponentials of the lower and upper score bounds to obtain L_A, U_A; let L_B, U_B describe its complement. Its mass lies in $[L_A/(L_A + U_B), U_A/(U_A + L_B)]$. Omitted candidates and temporal confirmation require separate treatment.

These conditional bounds are mathematical checks, not an enabled selection rule. They require inexpensive valid

prediction-error bounds, which the trajectory experiment does not provide. Neither measured RMSE nor a coarse–fine discrepancy is such a certificate. Positive-weight quadrature propagates score intervals for the same quadrature nodes, but integration error still needs its own bound.

APPENDIX V RECOVERY UNCERTAINTY, SWITCH TIMING AND ALARM CONTROL

A. UNCERTAINTY IN THE PRECEDING LOAD

After subtracting the mean recovery trajectory, consider a stacked report vector $r = s_c a + uz + \varepsilon$. Here s_c is the new load signature, u is the residual trajectory per unit amplitude of a preceding load, $z \sim \mathcal{N}(0, \sigma_z^2)$, and $\varepsilon \sim \mathcal{N}(0, R)$ is independent of z. Let $WRW^\top = I$ and $v = \sigma_z Wu$. For any difference d between candidate means, the squared distance under the common marginalized covariance follows from the rank-one inverse identity:

$$\|d\|^2_{(R+\sigma_z^2 uu^\top)^{-1}} = \|Wd\|^2 - \frac{(v^\top Wd)^2}{1 + v^\top v}. \tag{62}$$

The loss depends on the alignment with recovery as seen by the PMUs. With several Gaussian nuisance directions the same construction uses $R + UKU^\top$. A covariance that varies with the candidate requires the corresponding quadratic and log-determinant terms for each likelihood.

We used the nominal bank operators, all 47 coordinates, and the common quiet-null covariance over 30 reports. Each of nine preceding loads lasts one second and restores using its OFF operator. Each of nine new loads takes either sign at 0.2% of its setpoint. The four gaps and two prior scales give 1 296 deterministic contrasts. Rivals include all other catalog assets. Load rivals fit their magnitude within $[-0.2, 0.2]$, and line rivals retain fixed signatures. The recovery direction is shared in each contrast, including against a line rival. The nearest separation shrinks by at most 8.4%; the nearest rival changes in 0 contrasts. Fig. 19 shows the distribution of the losses. The tested covariance represents one uncertain amplitude, so the result does not cover uncertainty in the recovery source, timing or plant parameters.

B. SENSITIVITY WITHIN A REPORT INTERVAL

For an affine mode flow with continuous differential state, shifting a prescribed switch at τ inserts the sensitivity $f_-(x_\tau) - f_+(x_\tau)$ at the switch. The subsequent variational flow propagates that difference [10]. For a line that opens and recloses, the onset derivative propagates through both modes. The restoration derivative starts at the second switch. The algebraic observation map is fixed on each open interval between reports and changes when a candidate switch crosses a report.

We checked both derivatives on the saved nominal and line 23–24 operators using the eight PMU voltages in rectangular coordinates. Four noiseless cases generated and fitted with the same model place both switches off the report grid. Central differences agree with the derivatives to a maximum relative error of 1.11×10^{-7}. Starting at the interval midpoints gives

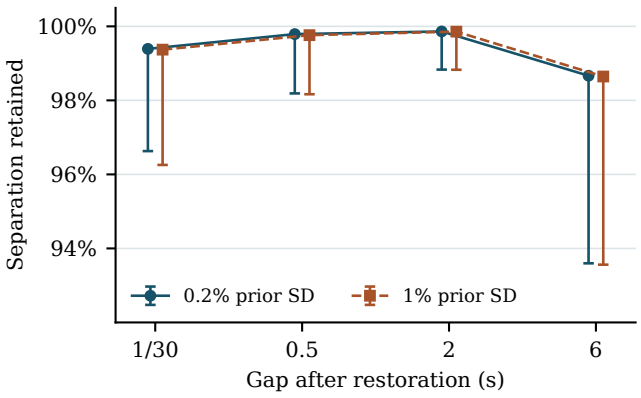


FIGURE 19. Nominal separation retained when the preceding load amplitude is uncertain. Markers show medians and lower whiskers show tenth percentiles over 162 deterministic contrasts per gap and prior scale. The horizontal positions represent four discrete gap settings.

errors up to 11.67 ms. Fitting within the supplied intervals recovers both switch times to within 0.01 ms in all four cases. Timing and asset search, model mismatch, and measurement noise remain outside this derivative check.

C. SEQUENTIAL EVIDENCE UNDER NULL MISMATCH

For normalized predictive densities q_t and $p_{0,t}$ selected from past reports, the product of $q_t(y_t)/p_{0,t}(y_t)$ is a nonnegative martingale under a correctly specified null with common support. Ville's inequality bounds the probability of ever crossing $1/\alpha$ by α for one test starting at one [11]. This probability refers to the full test and differs from the published alarm count per hour.

We illustrate the dependence on the null in 20,000 simulated sequences of 120 independent Gaussian reports. The numerator is $\mathcal{N}(0.5, 1)$ and the denominator is $\mathcal{N}(0, 1)$. At $\alpha = 0.05$, 3.57% of matched-null sequences cross the threshold. With the same noise draws and an unmodeled 0.35 shift of the actual null mean, 85.66% cross. The shift changes the expected log increment from -0.125 to 0.05. The example motivates checking the normal predictive law before using a sequential guarantee for PMU monitoring.

APPENDIX W MEMORY COMPRESSION AND STARTUP COVARIANCE

A. FOUR MATCHED REPRESENTATIONS OF RECOVERY

An external controlled example uses three swing nodes, a grounded reference, two possible recovery components and three measured coordinates. It compares a state mean, a moment-matched Gaussian, the full state mixture and the same mixture with historical cause labels. All arms use the same linear dynamics and class priors. The next load can occur at either of two sources at a known onset. Each arm uses 3,000 independent normal windows to set a 5% window false-positive target. Each scenario then evaluates 3,800 new windows. This target concerns fixed windows, whereas the article calibrates alarms per hour of continuous monitoring.

In the nominal colored-noise scenario, the mean, Gaussian and mixture correctly detect and locate 25%, 43.5% and 45.1% of event windows, respectively. Across the four scenarios, the mixture and labeled histories have identical new-source posteriors. This equality follows from the shared likelihoods and transition law after summing over component labels. Labels support queries about the previous cause. The recursive Gaussian calculation agrees with the batch calculation on every prefix of the checked trajectory. These results support the distinction in the predictive sufficiency analysis. The numerical gains refer to the small network and its prescribed recovery distribution.

B. ACCURACY OF A STARTUP COVARIANCE

For true and estimated positive definite covariances Σ and $\widehat{\Sigma}$, let $\eta = \|\Sigma^{-1/2}\widehat{\Sigma}\Sigma^{-1/2} - I\|_2$. When $\eta < 1$, every fixed signature d satisfies

$$\frac{d^\top \Sigma^{-1} d}{1+\eta} \le d^\top \widehat{\Sigma}^{-1} d \le \frac{d^\top \Sigma^{-1} d}{1-\eta}. \tag{63}$$

This follows by inverting the eigenvalue bounds in whitened coordinates. The inequality controls a fixed quadratic. A data-driven alarm rule also depends on log determinants, selection and the normal predictive law. The observed TVE is on a signal scale, whereas η measures relative covariance error.

We generated 128 Gaussian prefixes per setting in 47 coordinates, using the nominal innovation correlation as the reference. The shifted condition adds unit diagonal variance in standardized coordinates. At 300 reports in that condition, median excess marginal log loss in nats per vector is 0.0049, 2.52 and 0.00671 for a scaled nominal covariance, a sample covariance and Ledoit–Wolf shrinkage, respectively, with independent draws. For an AR(1) coefficient of 0.9 the corresponding losses are 0.0391, 42 and 20.4. Shrinkage improves on the unstructured sample estimate in these settings. The nominal correlation remains a useful prior in this construction. Recovering load attribution at physical independent PMU TVE requires testing the covariance in the full bank with independent calibration and evaluation.

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REFERENCES

[1] S. Särkkä, *Bayesian Filtering and Smoothing*. Cambridge, U.K.: Cambridge University Press, 2013.
[2] S. Fiaz, D. Zonetti, R. Ortega, J. M. A. Scherpen, and A. J. van der Schaft, "A port-hamiltonian approach to power network modeling and analysis," *European Journal of Control*, vol. 19, no. 6, pp. 477–485, 2013.
[3] M. A. Pai, *Energy Function Analysis for Power System Stability*. Boston, MA, USA: Kluwer Academic Publishers, 1989.
[4] E. Y. Chow and A. S. Willsky, "Analytical redundancy and the design of robust failure detection systems," *IEEE Transactions on Automatic Control*, vol. 29, no. 7, pp. 603–614, 1984.
[5] A. Abur and A. Gómez Expósito, *Power System State Estimation: Theory and Implementation*. New York, NY, USA: Marcel Dekker, 2004.
[6] D. Eriksson, E. Frisk, and M. Krysander, "A method for quantitative fault diagnosability analysis of stochastic linear descriptor models," *Automatica*, vol. 49, no. 6, pp. 1591–1600, 2013.
[7] L. L. Scharf and B. Friedlander, "Matched subspace detectors," *IEEE Transactions on Signal Processing*, vol. 42, no. 8, pp. 2146–2157, 1994.
[8] Y. Wang, Y. Sun, V. Dinavahi, K. Wang, and D. Nan, "Robust dynamic state estimation of power systems with model uncertainties based on adaptive unscented H_∞ filter," *IET Generation, Transmission & Distribution*, vol. 13, no. 12, pp. 2455–2463, 2019.
[9] M. Katanic, J. Lygeros, and G. Hug, "Recursive dynamic state estimation for power systems with an incomplete nonlinear DAE model," *IET Generation, Transmission & Distribution*, vol. 18, no. 22, pp. 3657–3668, 2024.
[10] I. A. Hiskens and M. A. Pai, "Trajectory sensitivity analysis of hybrid systems," *IEEE Transactions on Circuits and Systems I: Fundamental Theory and Applications*, vol. 47, no. 2, pp. 204–220, 2000.
[11] S. R. Howard, A. Ramdas, J. McAuliffe, and J. Sekhon, "Time-uniform Chernoff bounds via nonnegative supermartingales," *Probability Surveys*, vol. 17, pp. 257–317, 2020.